

Unit-I: Introduction

Contents: Introduction to communication Theory - Basics of data communication, Types of Signals, A/D, D/A, A/A, D/D Signal Conversion Methods, Bandwidth Utilization and Data Rate Limits, Multiplexing Techniques, Data rate limits, Topologies, Noise, types of noise, Shannon Hartley Theorem, Channel capacity, Nyquist and Shannon Theorem, Bandwidth S/N trade off. Network Models And addressing - OSI Model TCP/IP Model (Data Format, Addressing Mechanisms, Devices).

Unit Objectives and outcomes: *On completion the students will be able to:*

1. To understand the fundamentals of communication system.
2. To understand the basics of internetworking.
3. To understand services and protocols used at Physical, Data Link, Network, Transport Layer.

Unit outcomes : On completion the students will be able to

CO1: Understand and explain the concepts of communication theory and compare functions of OSI and TCP/IP model. CO2: Analyze data link layer services, error detection and correction, linear block codes, cyclic codes, framing and flow control protocols.

CO3: Compare different access techniques, channelization and Ethernet standards.

CO4: Apply the skills of subnetting, supernetting and routing mechanisms.

CO5: Compare IPv4 and IPv6

CO6: Understand services and protocols used at transport layer.

Outcome Mapping:

PEOs: POs: COs: 1 PSOs:

Books used

1. Simon Haykin, "Communication Systems", 3E

2. Ranjan Bose, —Information Theory coding and Cryptography, McGraw-Hill, 2nd Ed

3. Bernard Sklar, —Digital Communication Fundamentals & applications, Pearson Education. Second Edition.

What is data?

Data (In computer science,)is anything in a form suitable for use with a computer. Data is often distinguished from programs

Data (generally) means groups of information that represent the qualitative or quantitative attributes of a variable or set of variables.

Data (plural of "datum", which is seldom used) are typically the results of measurements .

Data communications are the exchange of data between two **devices** via some form of transmission medium such as a wire cable.

Data & Data Communication Function

1. Delivery.

The system must deliver data to the correct destination. Data must be received by the intended device or user only.

2. Accuracy. :-The system must deliver the data accurately. Data that have haltered in transmission and left uncorrected are unusable.

3. Timeliness.

The system must deliver data in a timely manner. Data delivered late useless. In the case of video and audio, timely delivery means delivering data they are produced, in the same order that they are produced, and without signal cant delay. This kind of delivery is called *real-time transmission*.

4. Jitter. Jitter refers to the variation in the packet arrival time. It is the uneven delay; the delivery of audio or video packets. For example, let us assume that video pad are sent every 30 ms. If some of the packets arrive with 30-ms delay and others v 40-ms delay, an uneven quality in the video is the result.

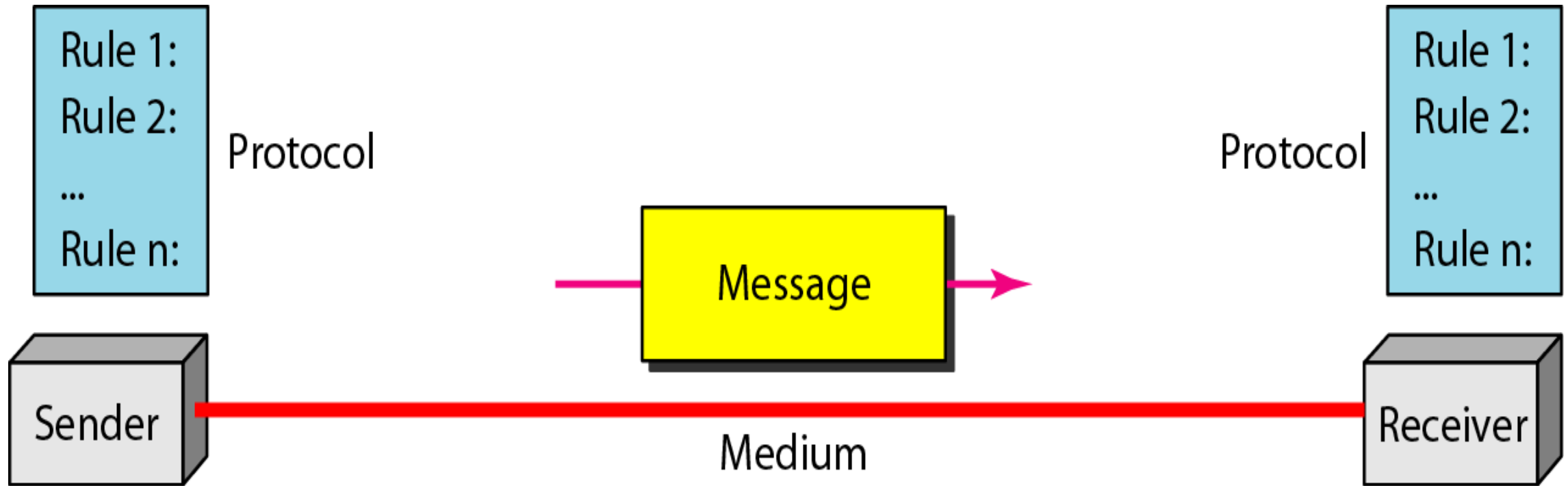
Data communication

- It is the engineering discipline concerned with communication between computer systems or devices.
- Networking, routers, routing protocols, and networking over the public Internet have their specifications defined in documents called RFCs.
- Computer networking is sometimes considered a sub-discipline of telecommunications, computer science, information technology and/or computer engineering. Computer networks rely heavily upon the theoretical and practical application of these scientific and engineering disciplines.

There are three types of networks: 1.Internet. 2.Intranet. 3.Extranet

- *internet* refer to the worldwide collection of networks that use open protocols such as TCP/IP to communicate with one another.
- *intranet* to refer to a communications network based on web technology but that's available only to certain people, such as the employees of a company.
- *extranet* refer to an extension of an intranet that uses internet protocols to give authorized outside users limited access to the intranet.

Five components of data communication



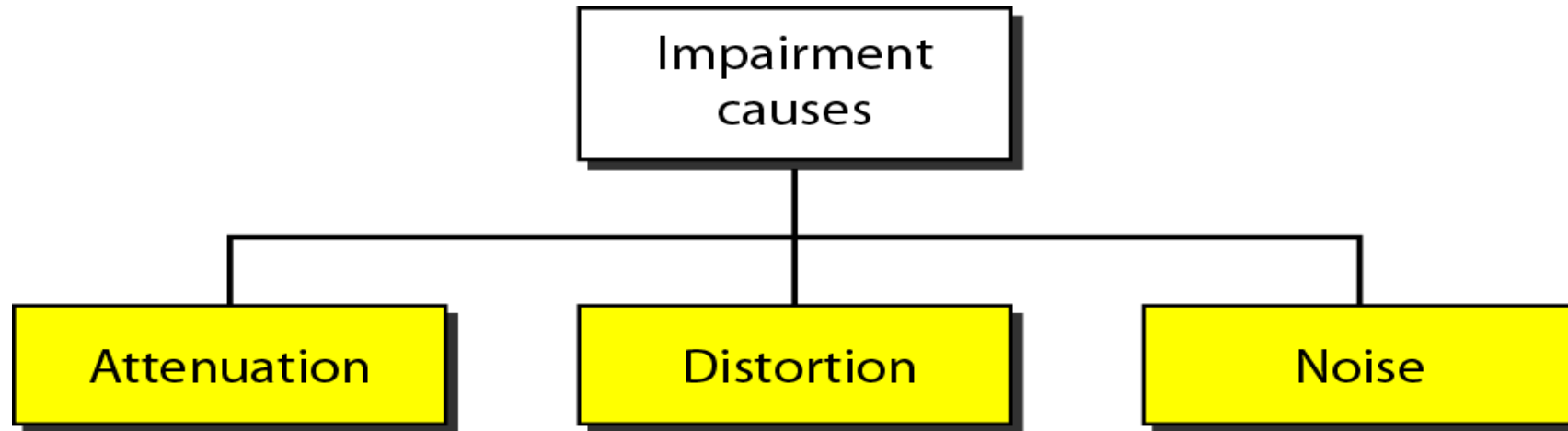
- Message : eg. Text, numbers, pictures, audio, video
- Sender/receiver: minimum two entities are required ;
eg: computers, telephone, TV etc
- Tx medium: eg. TP, Coax, FOC, radio waves

Impairment.

Transmission Impairments.

The signal received may differ from the signal transmitted. The effect will degrade the signal quality for analog signals and introduce bit errors for digital signals. There are three types of transmission impairments: attenuation, delay distortion, and noise.

Three causes of impairment are attenuation, distortion, and noise



Causes of impairment

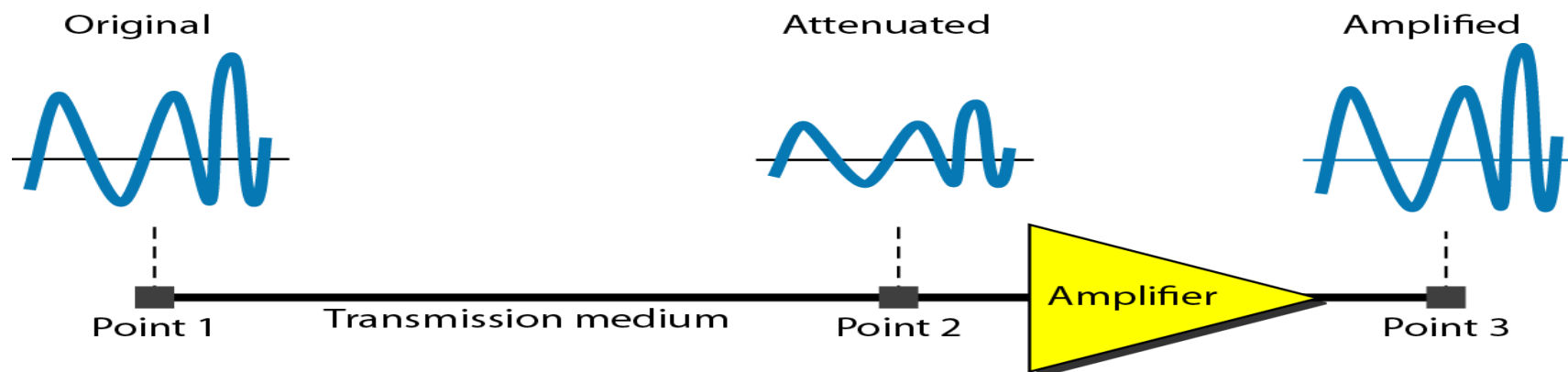
Attenuation:

It means loss of energy. The strength of signal decreases with increasing distance which causes loss of energy in overcoming resistance of medium. This is also known as attenuated signal.

Amplifiers are used to amplify the attenuated signal which gives the original signal back.

Attenuation is measured in decibels(dB). It measures the relative strengths of two signals or one signal at two different point.

Attenuation(dB) = $10\log_{10}(P2/P1)$, P1 is power at sending end and P2 is power at receiving end.



Causes of impairment

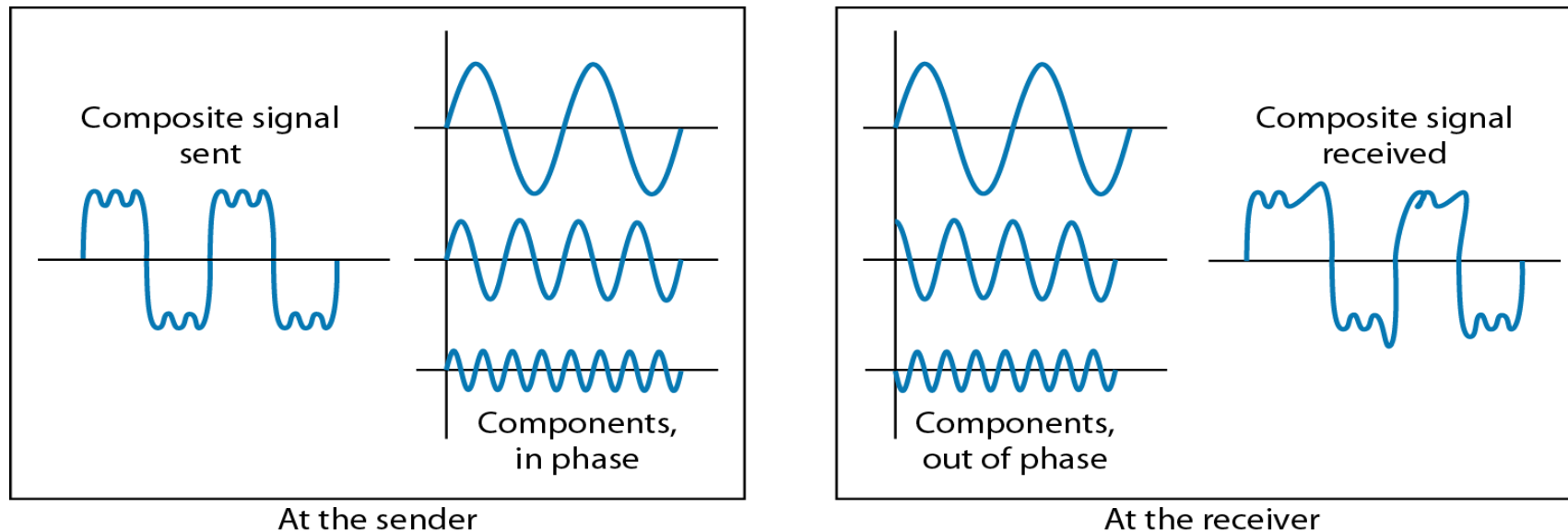
Distortion:

It means change in the shape of signal. This is generally seen in composite signals with different frequencies.

Each frequency component has its own propagation speed travelling through a medium.

Every component arrive at different time which leads to delay distortion.

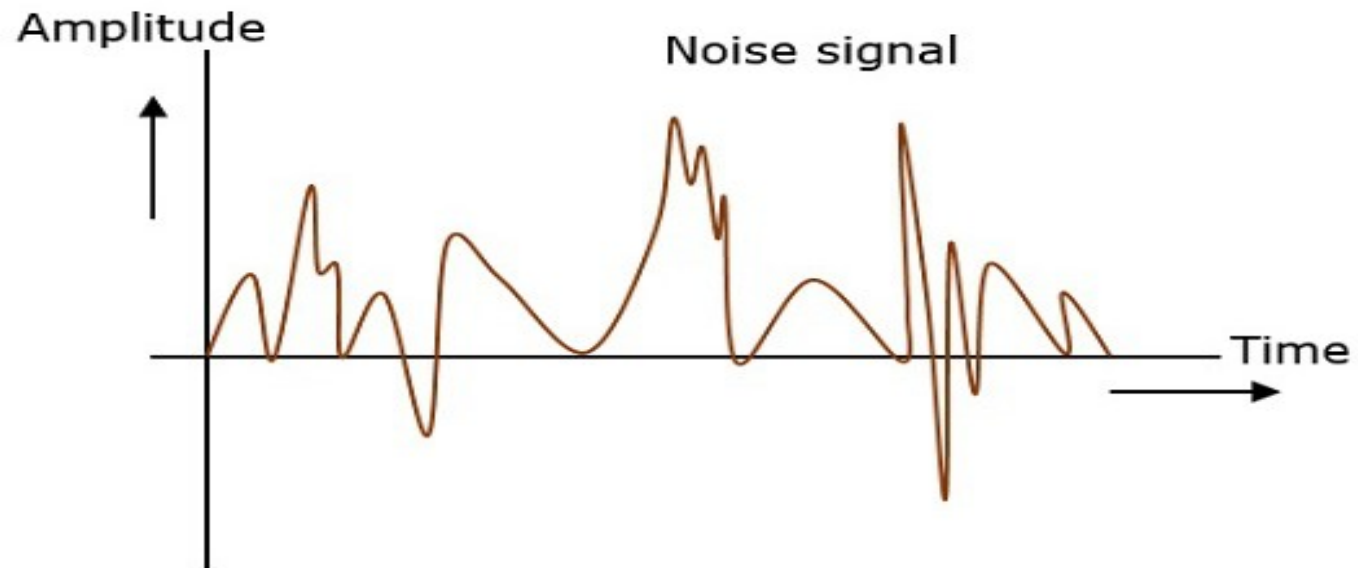
Therefore, they have different phases at receiver end from what they had at senders end.



Noise

Noise

Noise is a unwanted electrical disturbance that gives rise to audible or visual disturbance in communication system and errors in digital communication. Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Hence, it is understood that noise is some signal which has no pattern and no constant frequency or amplitude. It is quite random and unpredictable. Measures are usually taken to reduce it, though it can't be completely eliminated.



Causes of Impairments

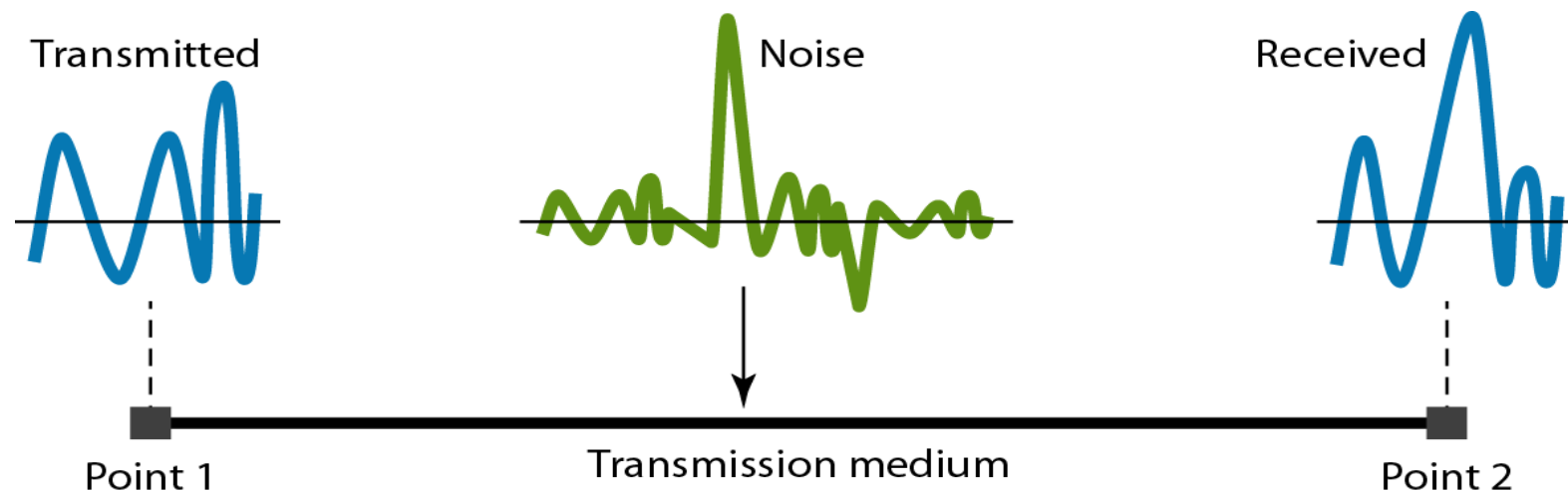
Noise:- The random or unwanted signal that mixes up with the original signal is called noise. There are several types of noise such as induced noise, crosstalk noise, thermal noise and impulse noise which may corrupt the signal.

Induced noise comes from sources such as motors and appliances. These devices act as sending antenna and transmission medium act as receiving antenna.

Thermal noise is movement of electrons in wire which creates an extra signal.

Crosstalk noise is when one wire affects the other wire.

Impulse noise is a signal with high energy that comes from lightning or power lines



Most common examples of noise are –

Hiss sound in radio receivers

Buzz sound in-between of telephone conversations

Flicker in television receivers, etc.



Effects of Noise

Noise limits the operating range of the systems

Noise indirectly places a limit on the weakest signal that can be amplified by an amplifier.

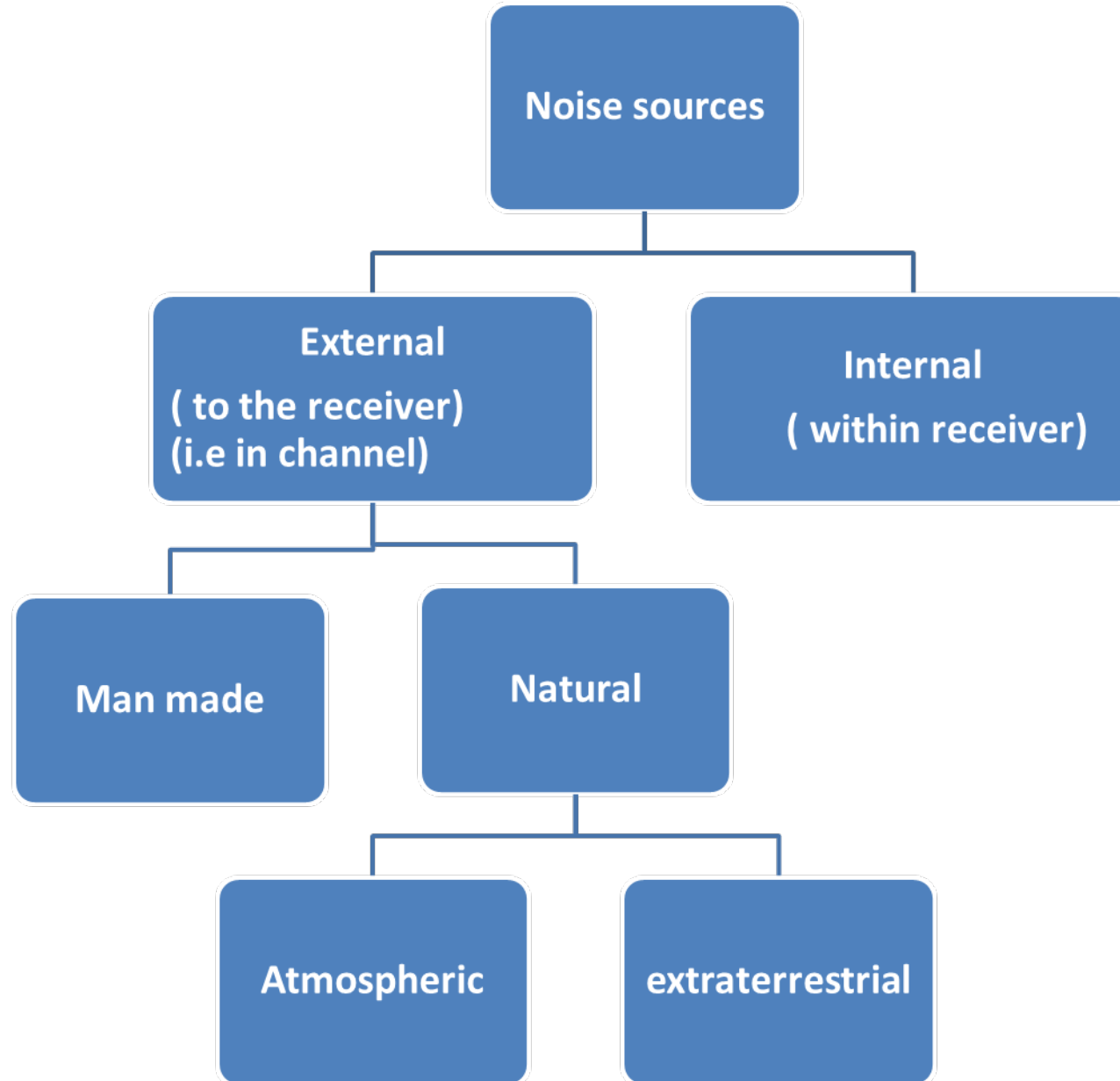
The oscillator in the mixer circuit may limit its frequency because of noise. A system's operation depends on the operation of its circuits. Noise limits the smallest signal that a receiver is capable of processing.

Noise affects the sensitivity of receivers

Sensitivity is the minimum amount of input signal necessary to obtain the specified quality output.

Noise affects the sensitivity of a receiver system, which eventually affects the output.

Noise



Natural sources of noise

Two types: atmospheric (static) and Extraterrestrial Noise

Atmospheric

is caused by lightning discharges in thunderstorms and other natural electric disturbances occurring in the atmosphere.

becomes less severe at frequencies above about 30 MHz because of the higher frequencies are limited to line-of-sight propagation i.e. less than 80 kilometers or so.

Extraterrestrial Noise

Two types: Solar Noise & Cosmic Noise

- **Solar Noise** :

Constant noise radiation from the sun, simply because it is a large body at a very high temperature (over 6000°C on the surface).

It therefore radiates over a very broad frequency spectrum which includes the frequencies we use for communication.

Also, the sun is a constantly changing star which undergoes cycles of peak activity from which electrical disturbances erupt, such as corona flares and sunspots.

Natural sources of noise

- **Cosmic Noise**

Since distant stars are also suns and have high temperatures, they radiate RF noise in the same manner as our sun. and although they are far but their number is high, so their noise can become significant.

The noise received is called thermal (or black-body) noise .

This noise also includes noise from our own galaxy (the Milky Way), "quasars" and "pulsars."

- Extraterrestrial Noise is observable at frequencies in the range from about 8 MHz to 1.43 GHz

Industrial Noise (Man made)

Occur in the range of frequencies of 1 to 600 MHz (in urban suburban and other industrial areas)

Eg. automobile and aircraft ignition, electric motors and switching equipment; leakage from high-voltage lines , other heavy electric machines & Fluorescent lights.

INTERNAL NOISE

It is noise created by any of the active(transistors) or passive (resistors)devices used in receivers

Types:

1. Thermal Agitation Noise
2. Shot Noise
3. Transit-Time Noise

Introduction To basics of networking

Computer network fundamentals

ISO OSI Model: All Layers

TCP/IP Protocol Suite: All Layers

addressing (Physical, Logical Port and Other)

LAN, WAN And MAN

Network Topologies

Advantage of network

A) Sharing of computing (H/W & S/W)and information resources

- **File sharing** – you can easily share data between different users, or access it remotely if you keep it on other connected devices.
- **Resource sharing** – using network-connected peripheral devices like printers, scanners and copiers, or sharing software between multiple users, saves money.
- **Sharing a single internet connection** – it is cost-efficient and can help protect your systems if you properly secure the network.
- **Increasing storage capacity** – you can access files and multimedia, such as images and music, which you store remotely on other machines or network-attached storage devices.

B) Storing information in one centralized database can also help you **reduce costs** and also get **more efficiency**

- staff can deal with more customers in less time, since they have shared access to customer and product databases
- you can centralize network administration, meaning less IT support is required
- you can cut costs through sharing of peripherals and internet access
- You can **reduce errors** and **improve consistency** by having all staff work from a single source of information.

What is a network?

A network is a set of devices (often referred to as nodes) connected by communication links.

A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

A computer network is any set of computers or devices connected to each other with the ability to exchange data. There are three types of networks:

1.Internet. 2.Intranet. 3.Extranet

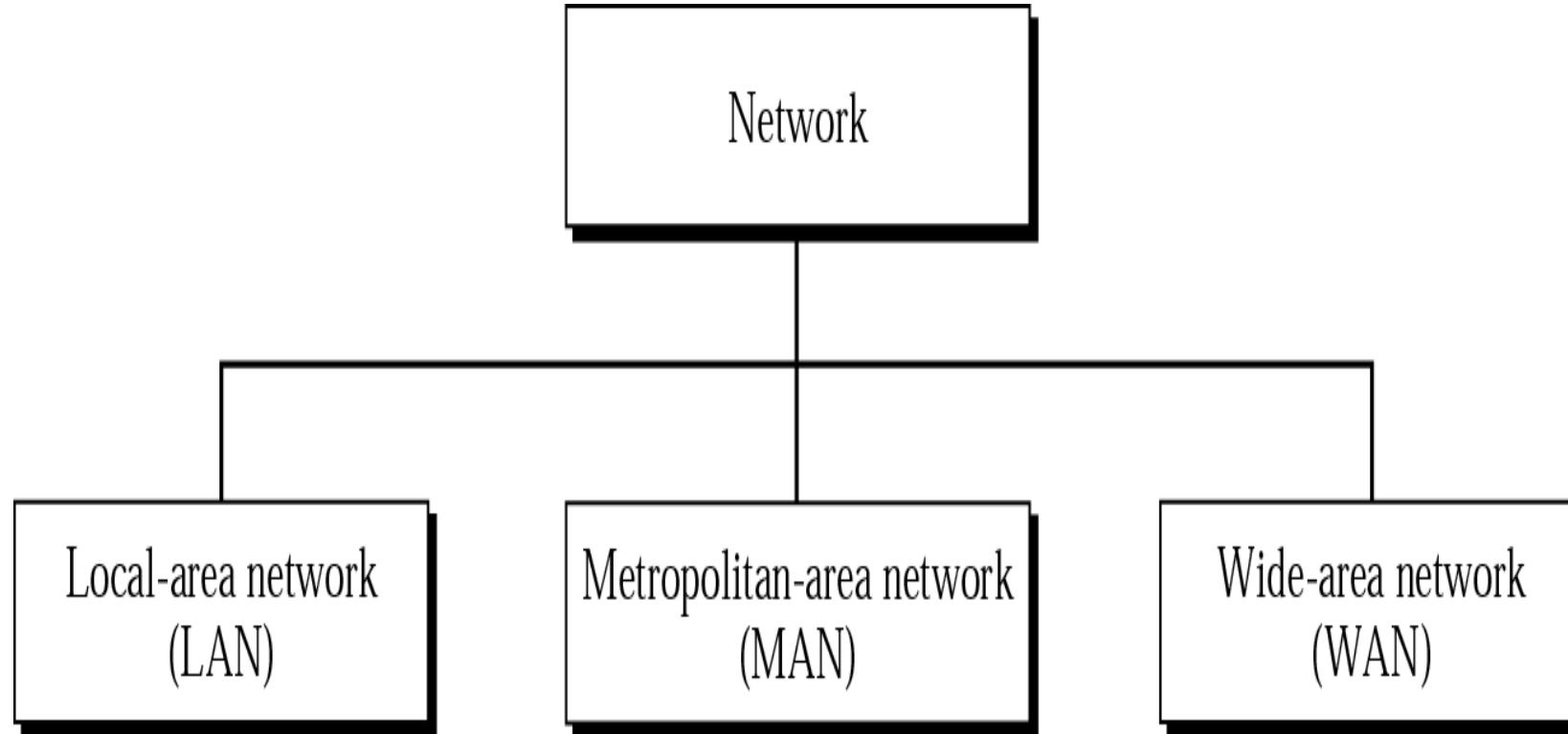
Examples of different networks are:

- Local area network (LAN),. An example of a LAN would be a computer network within a building.
- Metropolitan area network (MAN), examples a city or a state.
- Wide area network (WAN)
- Other types of CNs are PAN,CAN,WLAN/WiFi , Wimax etc

Networks are compared based on:

- **Performance:** in terms of transit time(time to reach destination) and response time(time between inquiry and response)is based on no. of users, type of Tx media, capability of H/W and efficiency of S/W. Performance is also measured by throughput and delay.
- **Reliability :** frequency of failures and ability to recover from failures.
- **Security :**

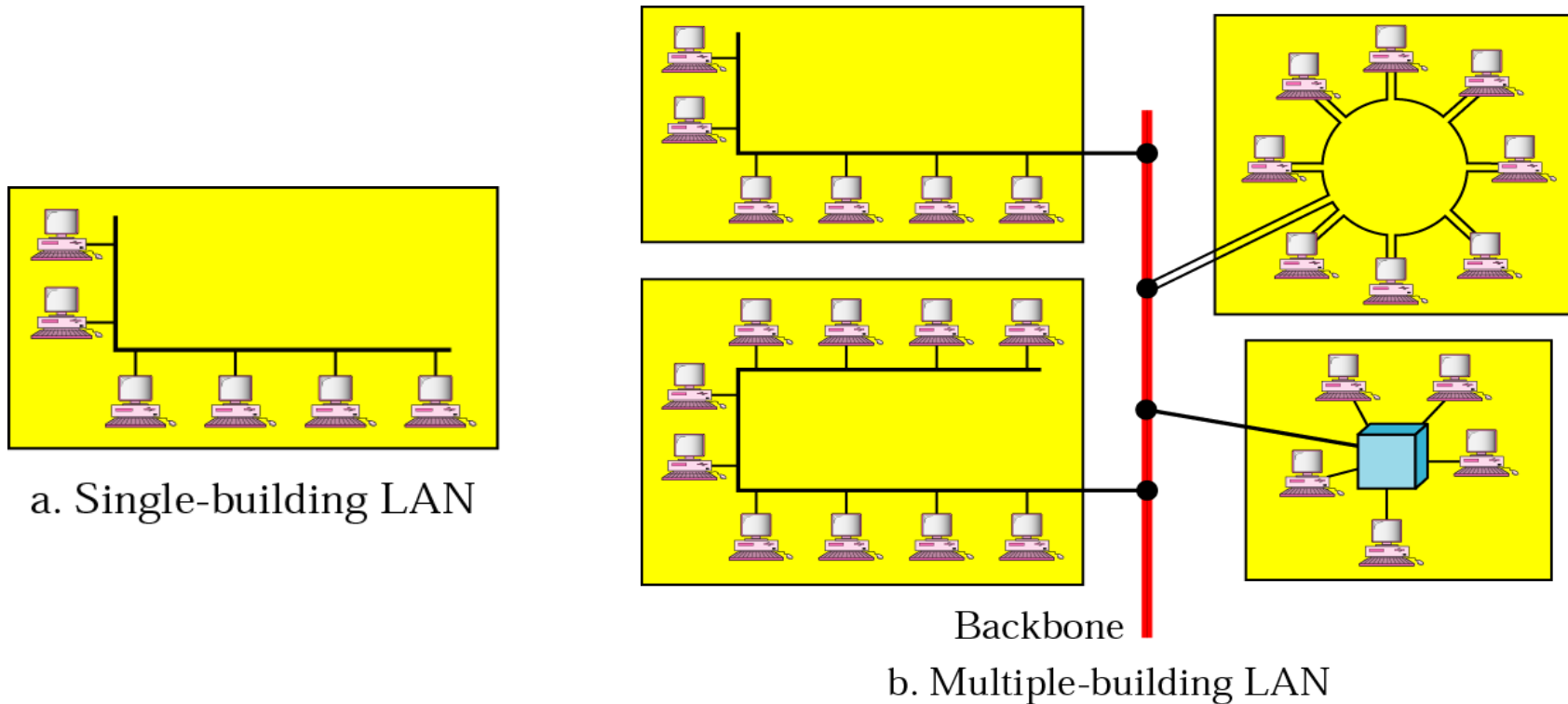
Categories of networks



- *other types are : CAN(campus) , PAN (personal) , BAN (body) , VAN (vehicular)*

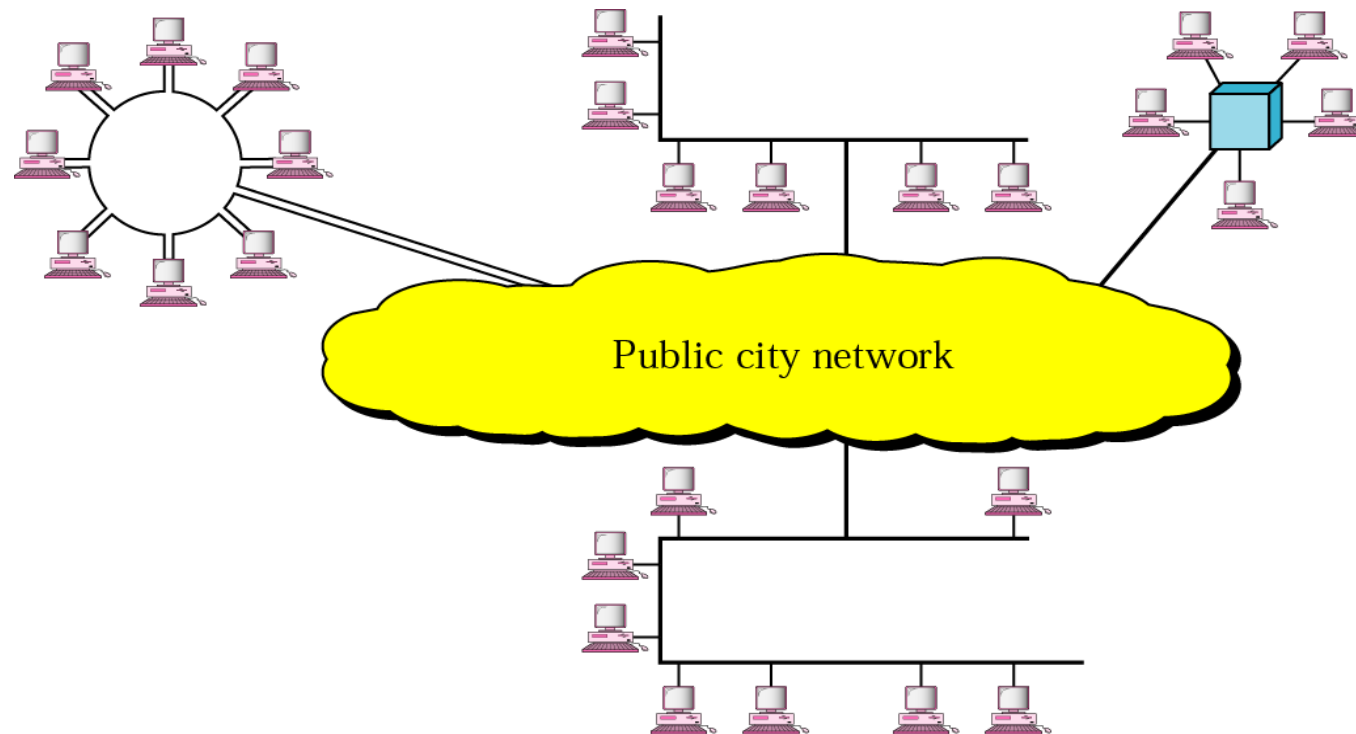
LAN :

usually privately owned and links the devices in a single Office ,building ,or campus. LAN size is limited to a few kilometers.



MAN

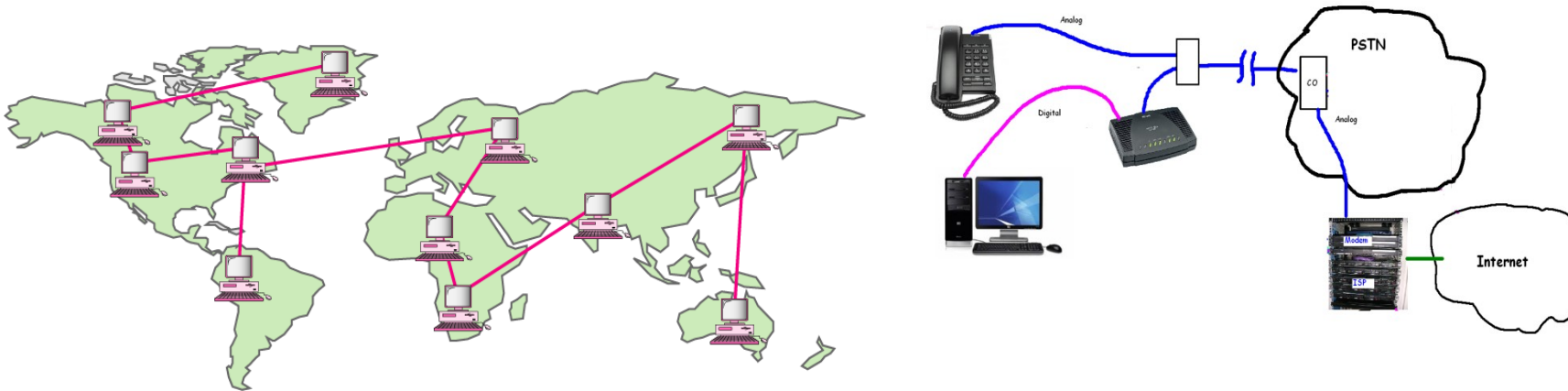
metropolitan (city)area networks: spans tens of miles



WAN

is worldwide

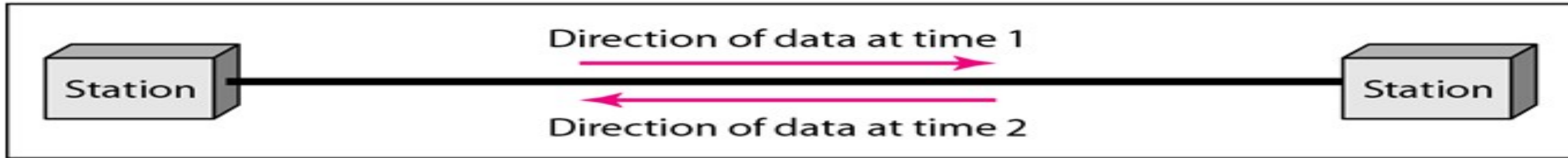
- A WAN can be as complex as the back bones that connect the Internet(called switched WAN)
- or as Simple as a dial-up line that connects a home computer to the Internet(called point to point WAN)



Types of Data flow in networks



a. Simplex



b. Half-duplex

Walky talky

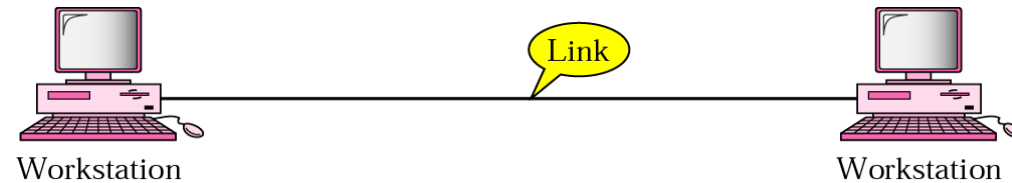


c. Full-duplex

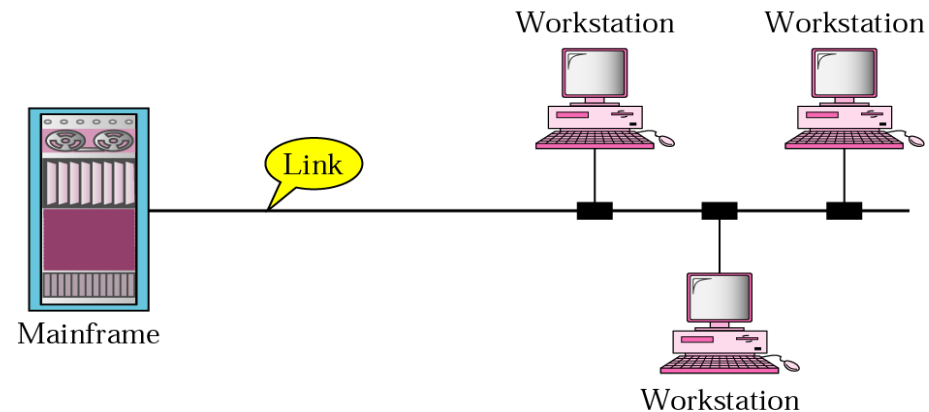
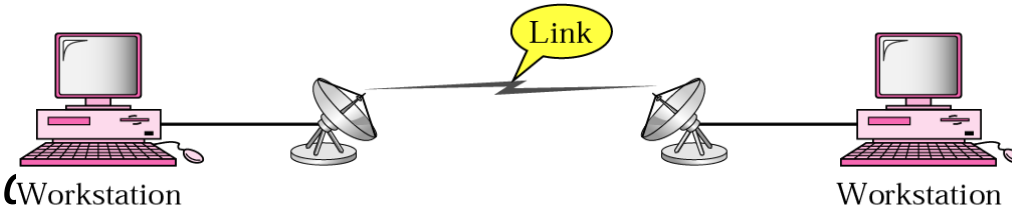
telephone

Type of connections:

- **1. Point-to-point connection** (dedicated link between two stations)



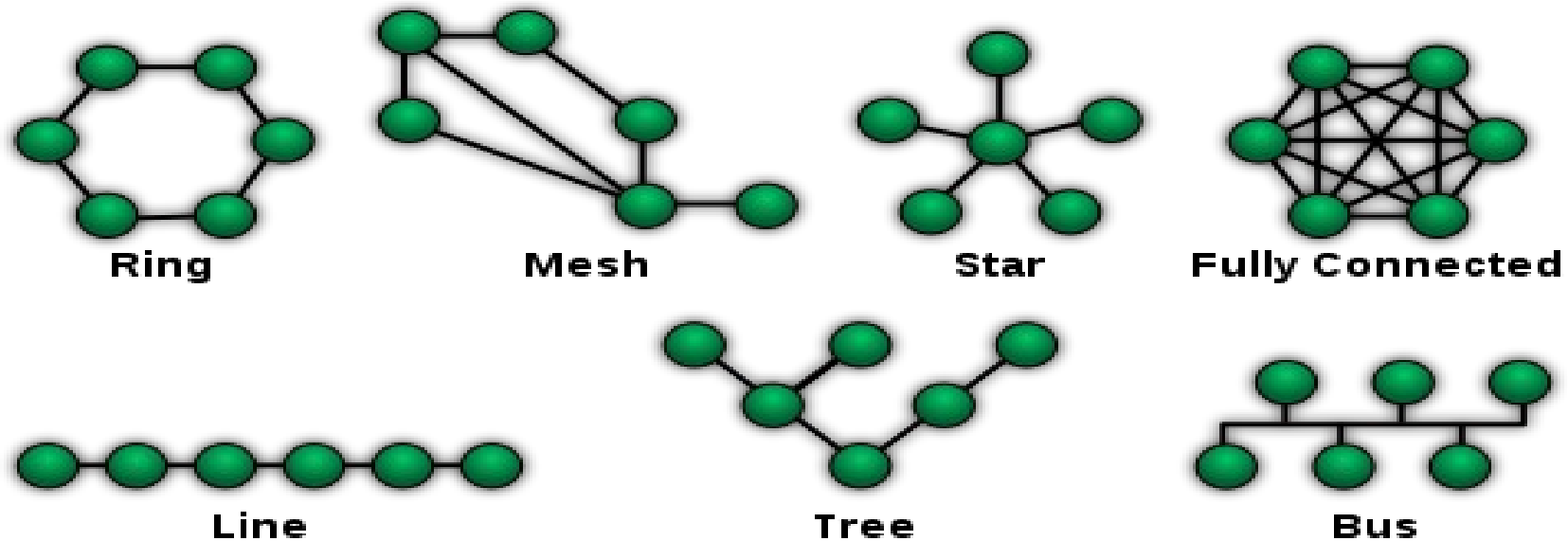
- **2. Multipoint connection** (shared link between multiple stations)



Topology

Topology (from the Greek topos: "place", and logis: "study") is the \ mathematical study of shapes. The arrangement of a network which comprises of nodes and connecting lines via sender and receiver is referred as network topology.

Physical design of network.



Two basic categories of network topologies :

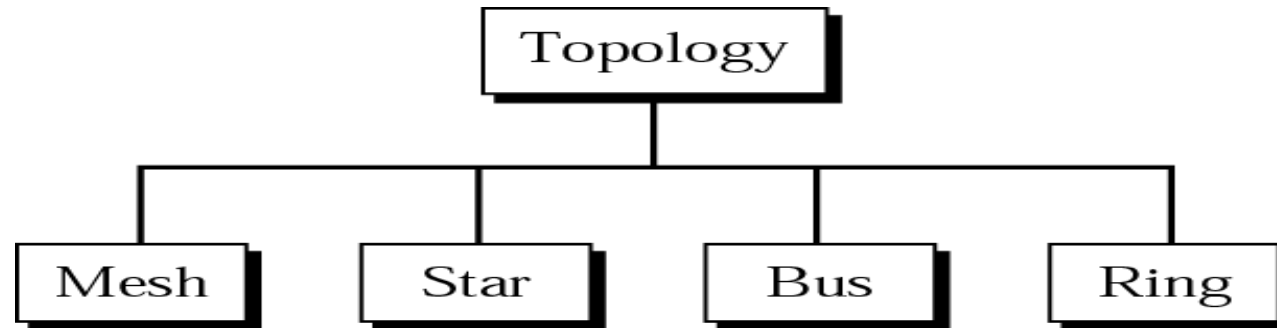
1) Physical topologies

2) Logical topologies.

Physical topology : Refers to the way in which a network is laid out physically.

The topology of a network is the geometric representation of the relationship of all the links and linking devices(usually called nodes)to one another.

Basic Categories of physical topology

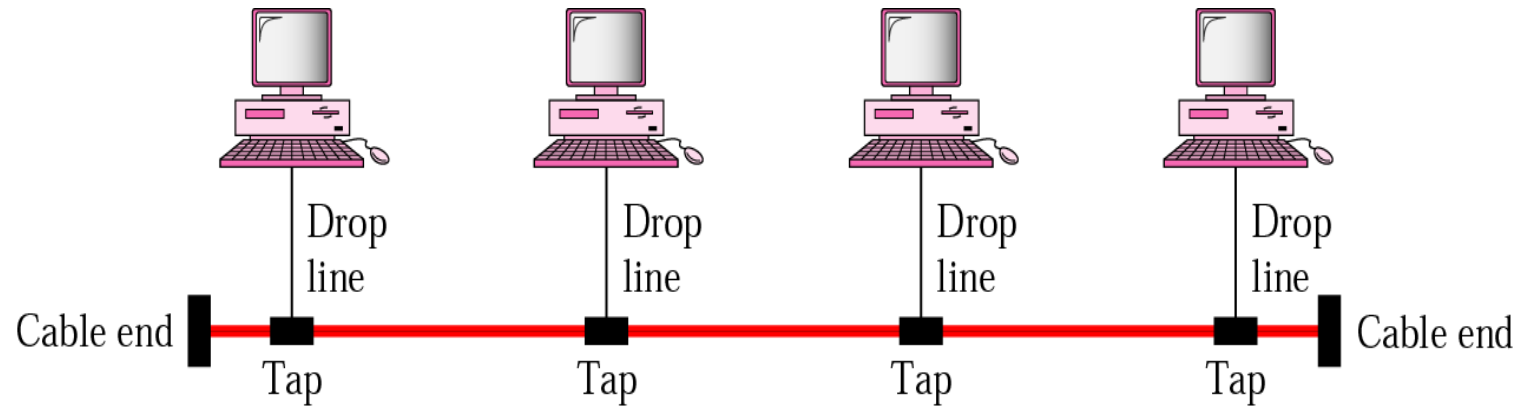


*The combination of these is called **hybrid topology** :*

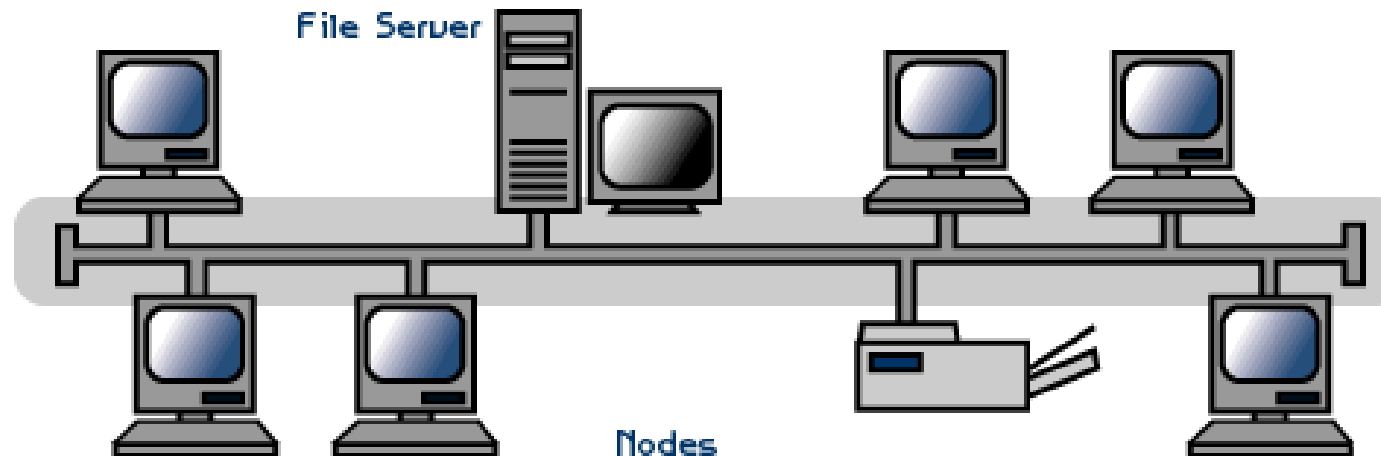
A network structure which is a combination of more than one topology is said to be hybrid topology.

Hybrid topology inherits merits and demerits of all the incorporating topologies

Bus Topology



- consists of a single cable with a terminator at each end.
- All nodes (file server, workstations, and peripherals) are connected to the cable via drop line and taps.



Bus Topology

Advantages of a Bus Topology

- Easy to connect a computer or peripheral to a bus.
- Requires less cable length than a star topology.

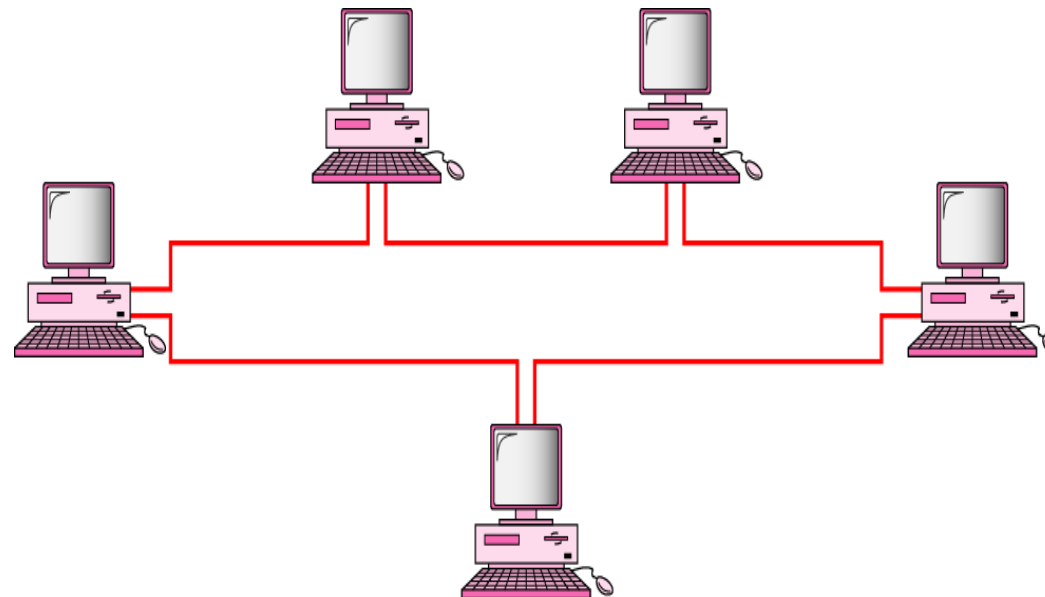
Disadvantages of a Bus Topology

- Entire network shuts down if there is a break in the main cable.
- Difficult reconnection and fault isolation : a fault or break in the bus cable stops all transmission, even between Devices on the same side of the problem.
- There is a limit on number of taps and distance between taps

Application: earlier version of ethernet LAN , but now less popular.

Ring Topology

- Each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction , from device to device , until it reaches its destination .
- A token is passed from one station to next. Only that station who has that token can send data. Each device has to wait its turn to get the token.



Ring Topology

Advantages

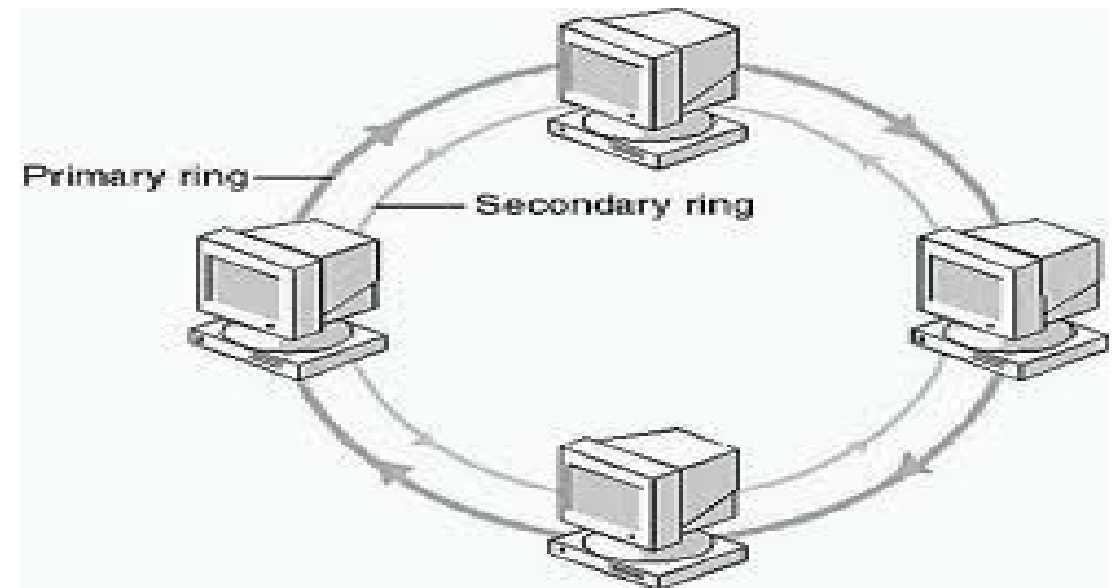
- **Easy to install and reconfigure** .Each device is linked to only its immediate neighbors(either physically or logically). To add or delete a device requires changing only two connections . The only constraints are maximum ring length and number of devices.
- **fault isolation is simplified**. a signal is circulating at all times. If one device does not receive a signal within a specified period ,it can issue an alarm.The alarm alerts the Network operator to the problem and its location.

Disadvantages : In a simple ring, a break in the ring (such as a disabled station)can disable the entire network. This

weakness can be solved by using a dual ring

Application :

IBMs token ring LAN , but less popular nowadays

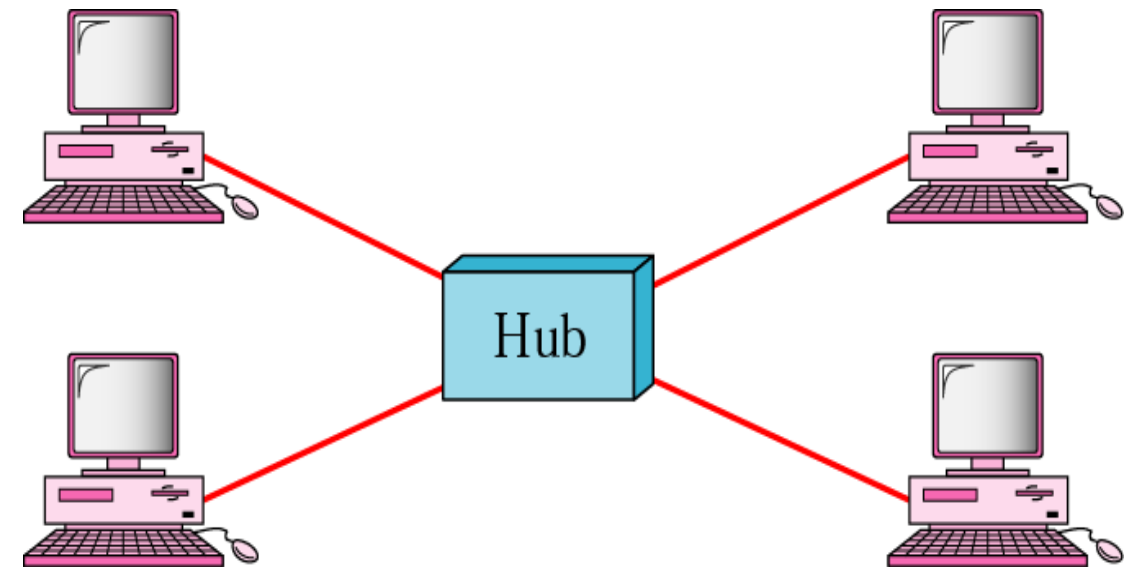
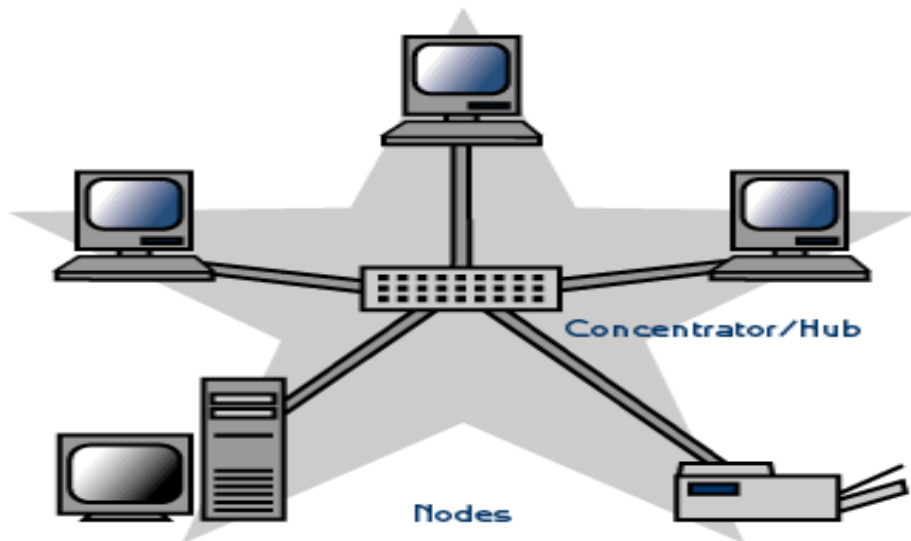


Star Topology :

Each device has a dedicated point-to-point link only to central controller, usually called a hub.

The devices are not directly linked to one another.(Unlike a mesh topology.)

The controller acts as an exchange :If one device wants to send data to another, it sends the data to the controller ,which then relays the data to the other connected device



Star Topology

Advantage:

- **easy to install and reconfigure:** each device needs only one link and one I/O port to connect it to any number of others.
- **Robustness.** If one link fails, only that link is affected .All Other links remain active. This factor also lends itself to easy fault identification & fault isolation.
- Hub can be used to monitor link problems and bypass defective links.

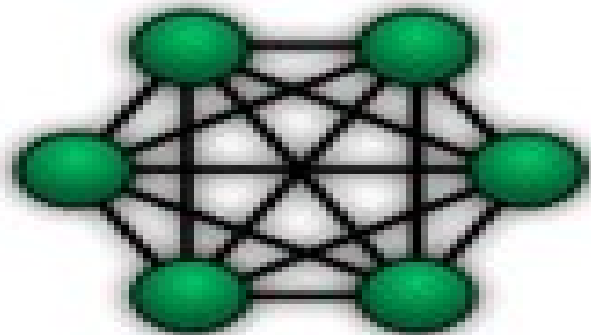
Disadvantage:

- **Cost :**less expensive than a mesh topology but more expensive than Bus
- the dependency of the whole topology on one single point , the hub. If the hub goes down ,the whole system is dead.
- **Cabling :**requires far less cable than a mesh, but more cabling is required than in some other topologies (such as ring or bus).

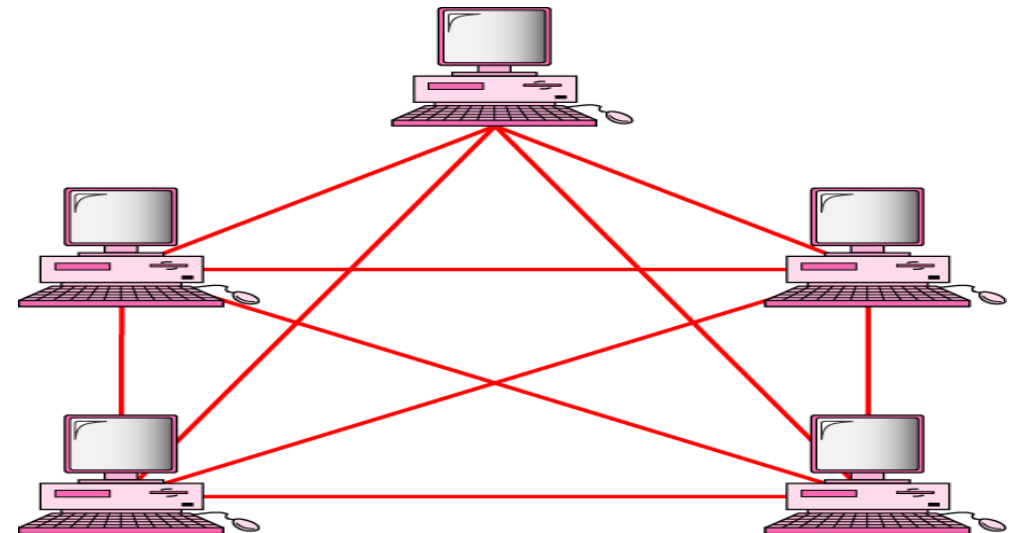
Application : local-area networks(LANs)

Full Mesh

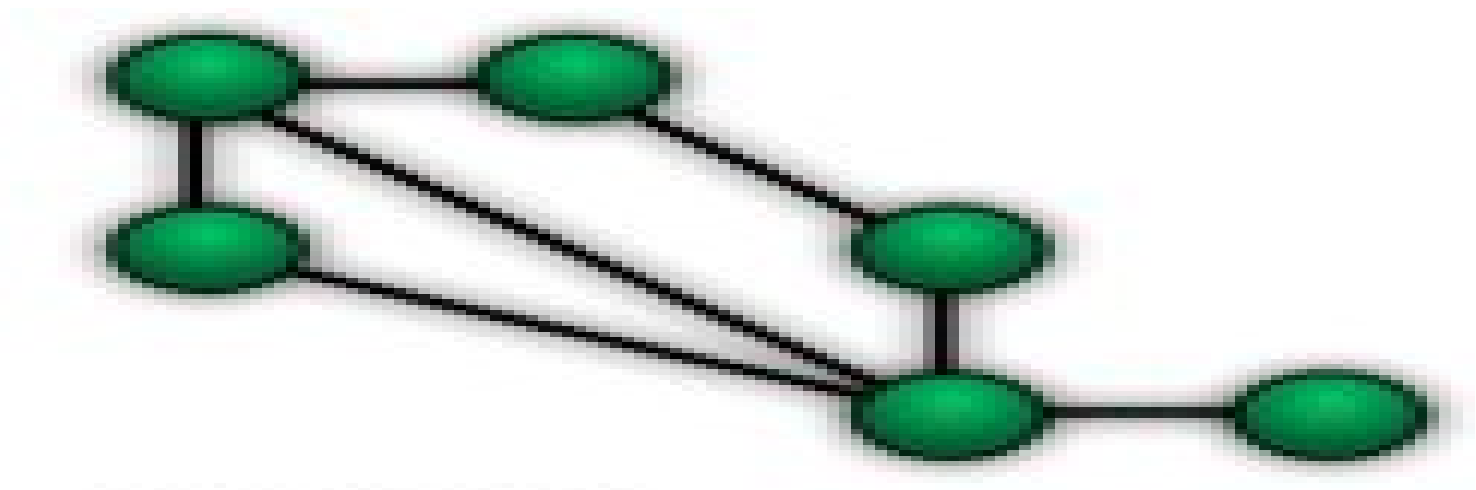
- In a mesh topology, every device has a dedicated point-to-point link to every other device .
- The term dedicated means that the link carries traffic only between the two devices it connects.
- The total number of links required to connect 'n' nodes is $\frac{n(n-1)}{2}$ and Each node has n-1 ports



Fully Connected Mesh



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Partial

Mesh Topology.

Advantages

- No problems of shared link (since link is not shared by any node other than the two nodes connected)
- **robust.** If one link becomes unusable ,it does not stop the entire system
- **Privacy or security**

Disadvantages

- Amount of cabling too bulky .
- Every device on the network must have $n-1$ input/output (iO)ports to be connected to the other $n-1$ stations
- So most expensive among all topologies

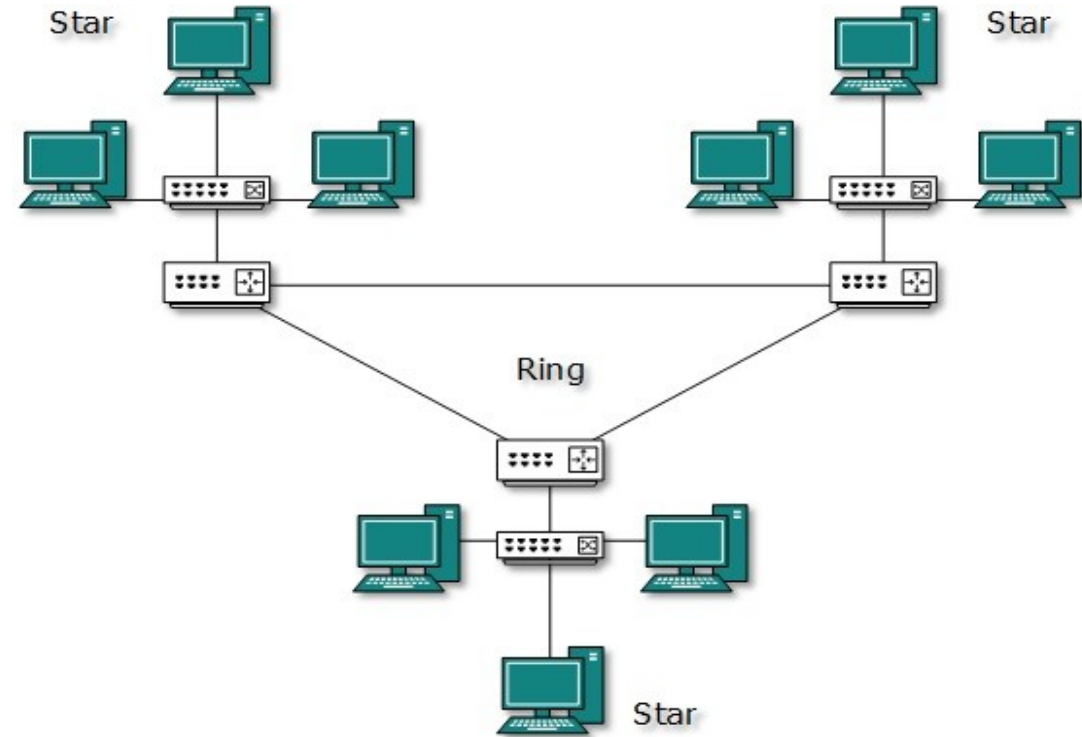
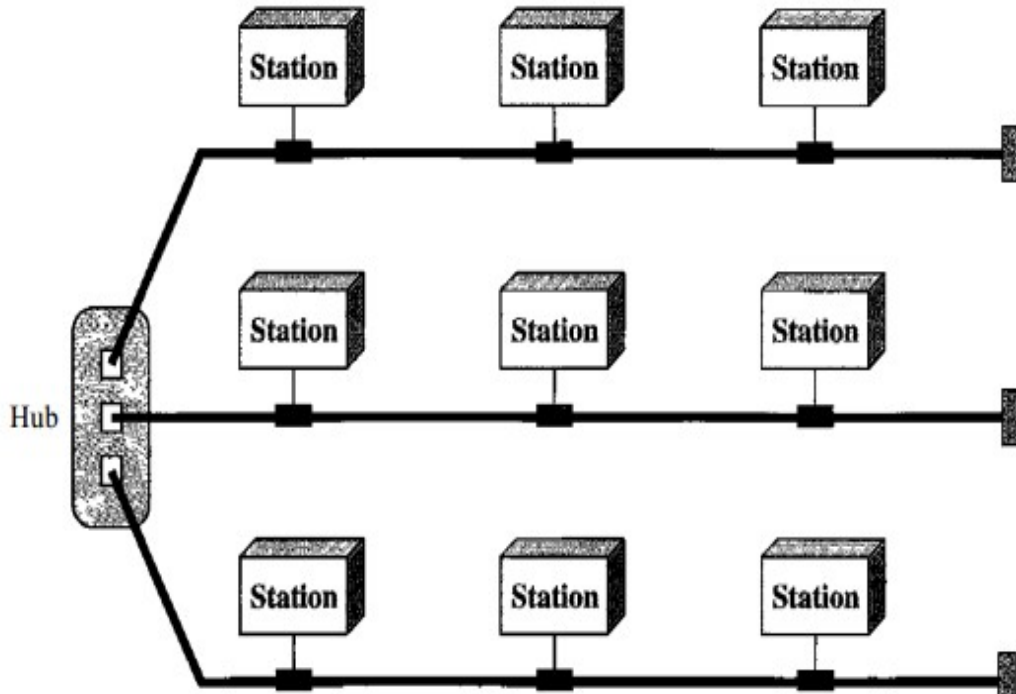
Application : to connect telephone exchanges together

- As a backbone connecting the main computers of other networks

Hybrid Topology :

A network structure whose design contains more than one topology is said to be hybrid topology.

Hybrid topology inherits merits and demerits of all the incorporating topologies



LAYERED TASKS

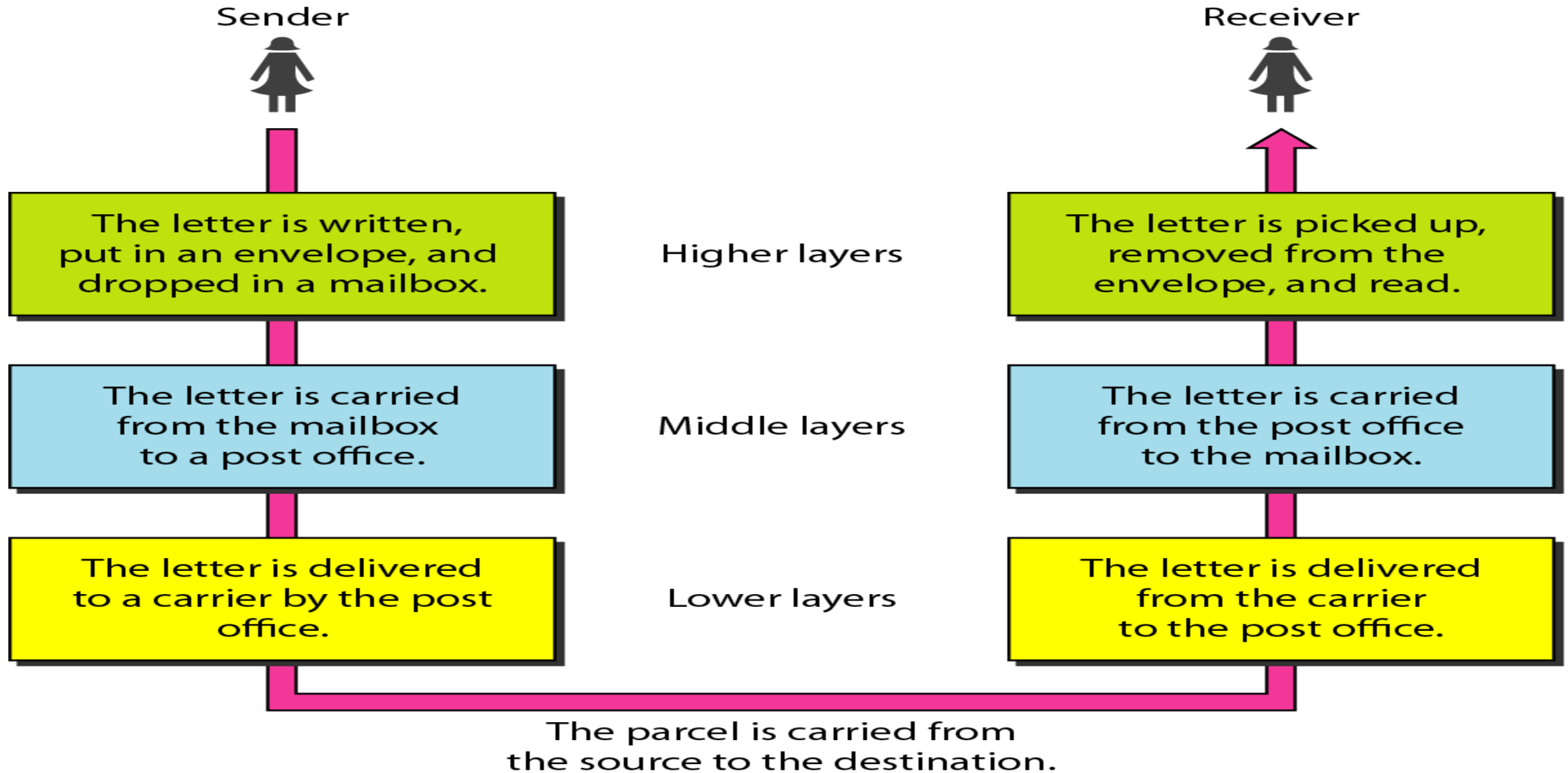
We use the concept of layers in our daily life.

As an example, let us consider two friends who communicate through postal mail.

The process of sending a letter to a friend would be complex if there were no services available from the post office.

Q. What are the tasks required to be performed for sending a letter ?

Tasks involved in sending a letter



THE ISO - OSI MODEL

This concept was used to model Data communication by OSI and TCP/IP. But OSI was never fully implemented.

And TCP/IP became more dominant since it was used extensively in internet before the OSI was created.

ISO (International Standards Organization)

- Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on various international standards.

THE ISO - OSI MODEL

OSI :

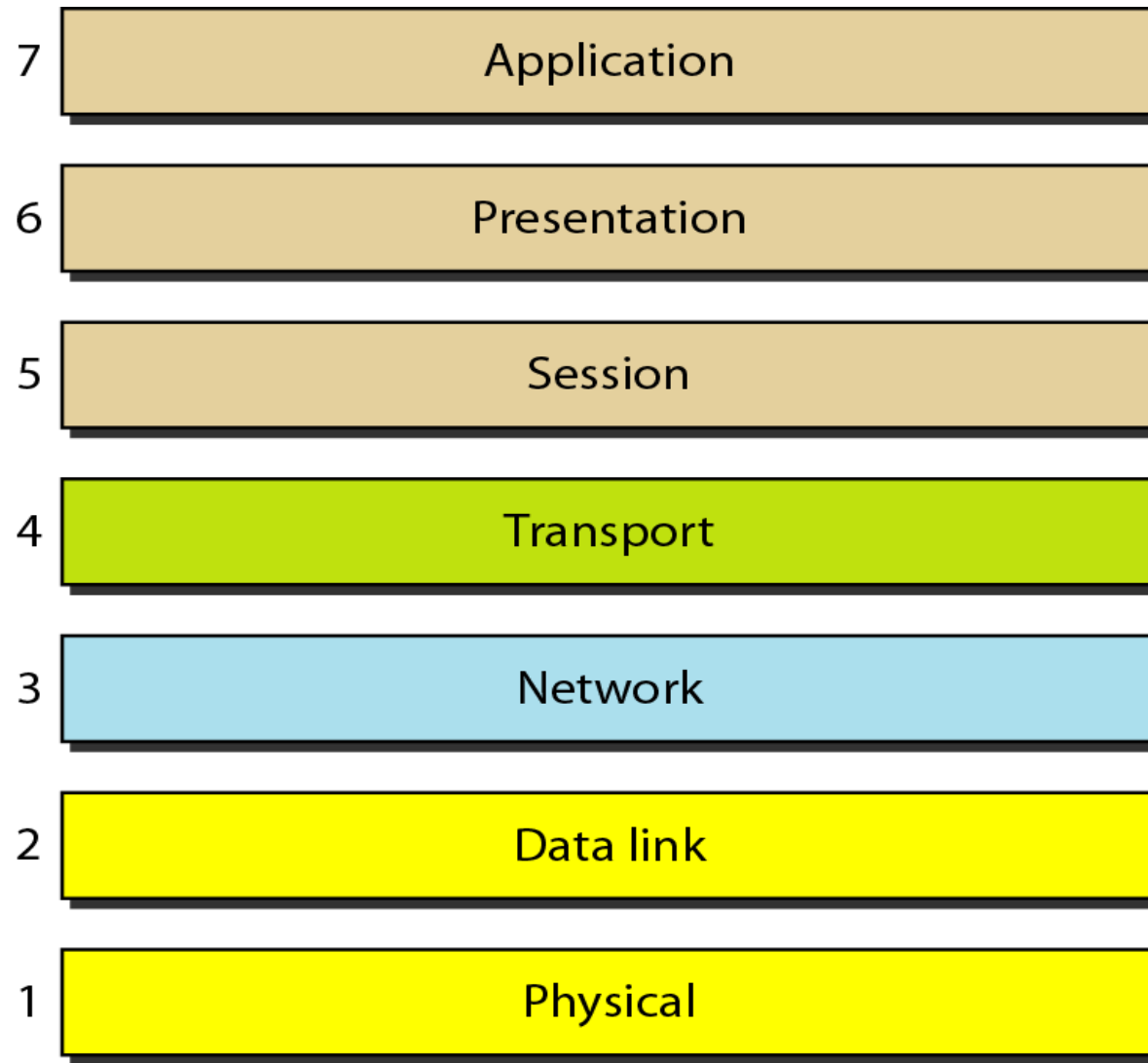
- An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model.
- It was first introduced in the 1977.
- Is an effort to standardize networking.
- Open system** is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture or changing it. i.e. interoperable
- OSI is **not protocol but a model** / framework for developing protocol standards.

The OSI model is used for two main reasons:

- **It divides the network communication process into smaller and simpler components(modules); thus aiding component development, design and troubleshooting.**
- **It provides a means by which changes in functionality in one layer does not require changes in other layers by Standardizing interfaces.**

i.e. It standardizes network components to allow multiple vendor development and support. i.e. It allows different types of network hardware and software to communicate with each other (interoperability).

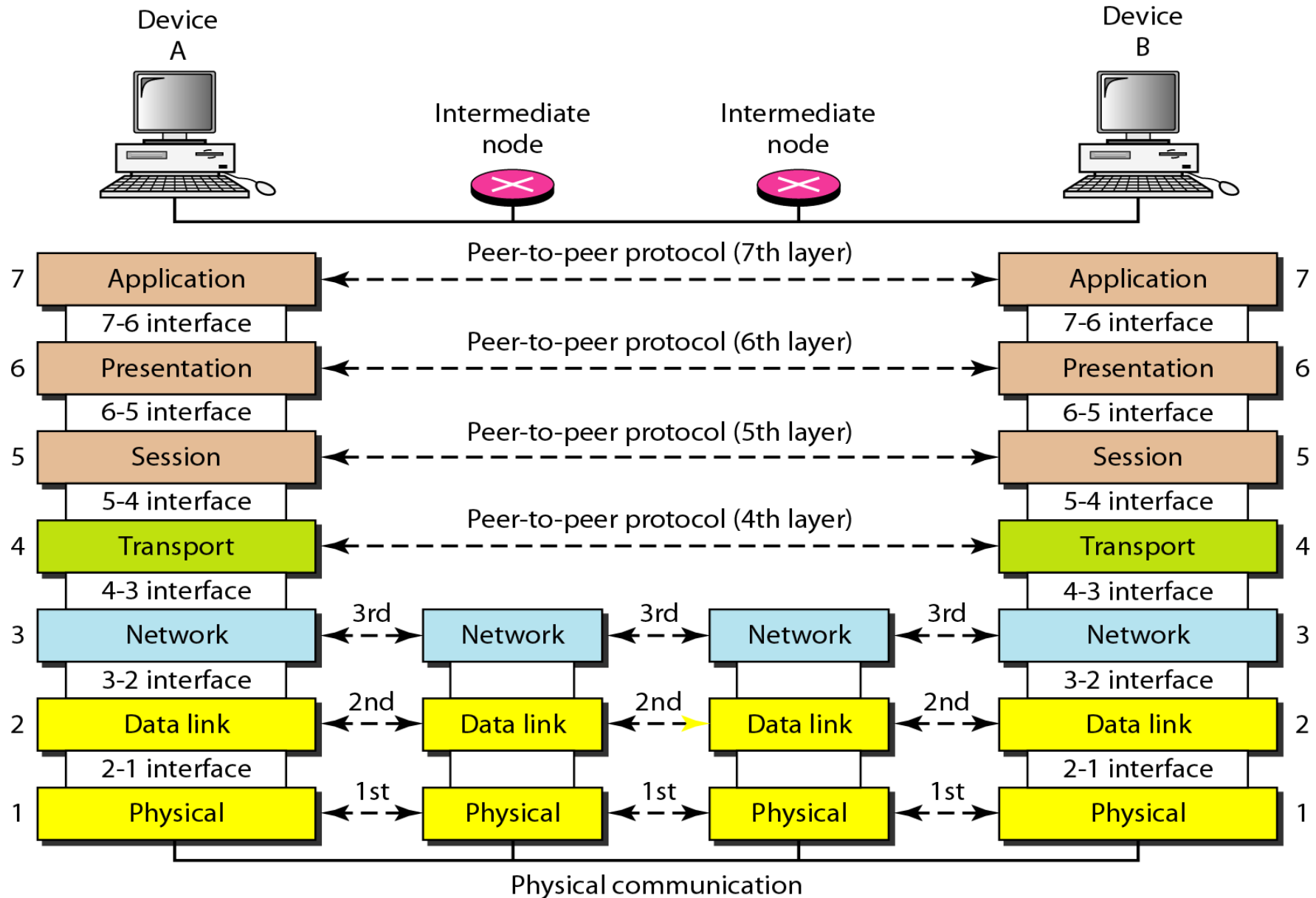
The OSI model



APSTNDP : All people seem to need data processing

- Each layer defines a set of functions distinct from other layers.
- Makes interoperability between other wise incompatible systems.
- Layer 1,2,3 : n/w support layers deals with physical moving of data from one device to next. Implemented as combination of H/W and S/W. except lowest physical which is implemented in H/W.
- Layers 5,6,7 : host or user support layers allows interoperability among unrelated s/w systems. Implemented in S/W
- Layer 4 links the two groups .

The interaction between layers in the OSI model



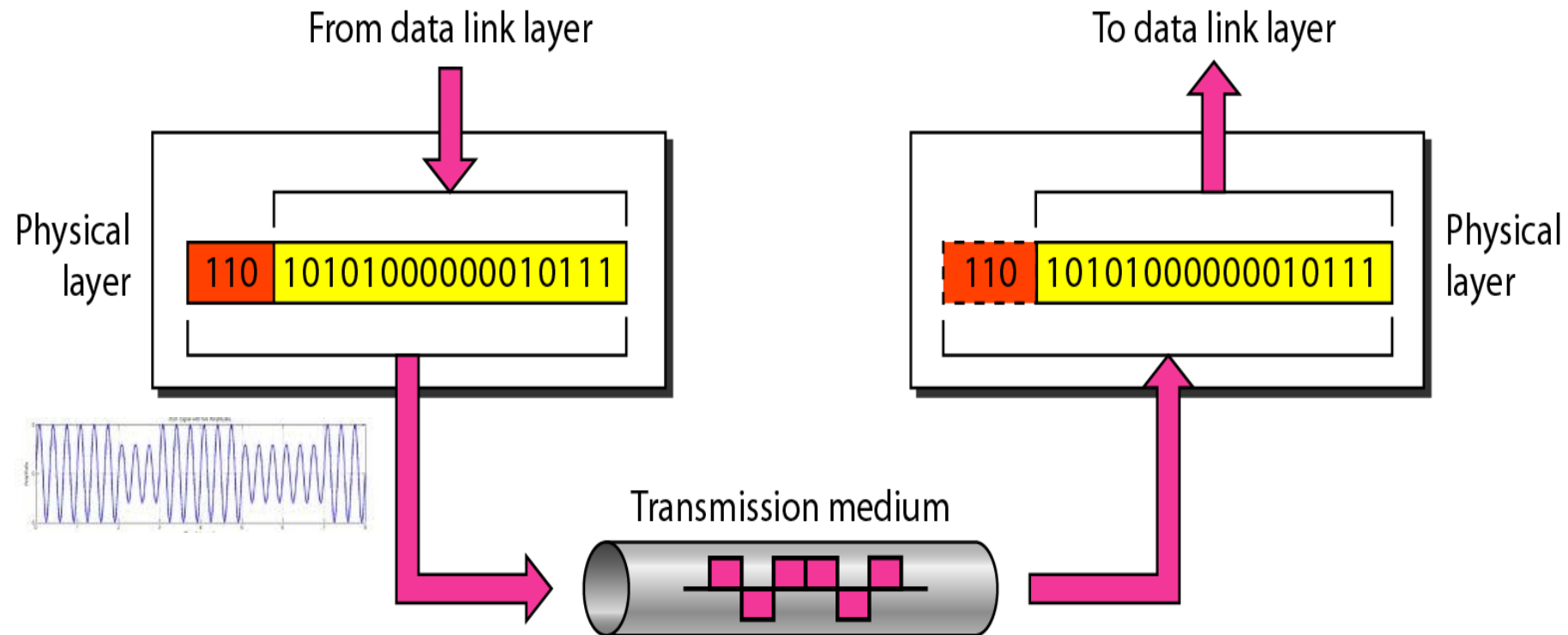
- A layer x on one system communicates with layer x on another system using agreed upon rules called protocols, and they are called **peer to peer protocols**.

Interface : list of services provided by lower layer to upper layer passing of data and related signaling between layers is done through this.

If interface is kept intact , the layers can be modified or replaced without changing the adjacent layers .

Physical layer

The physical layer is responsible for movements of individual bits(in form of signals) from one node to the next.

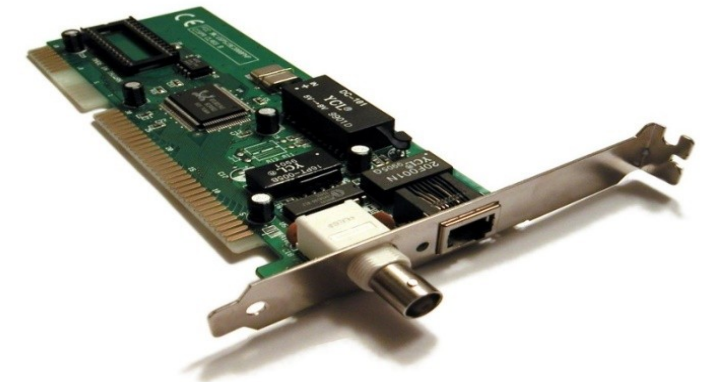
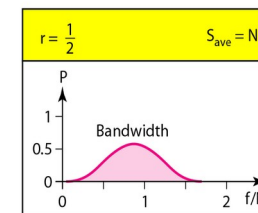
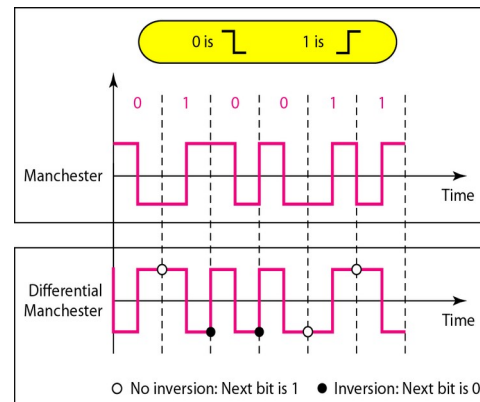
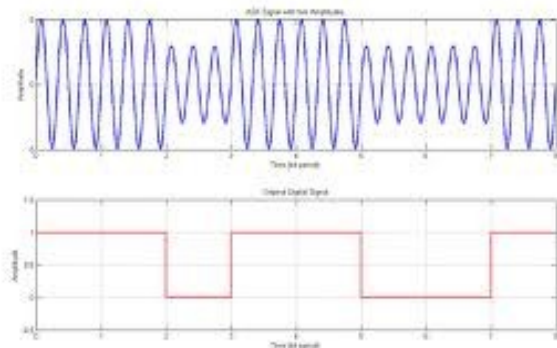
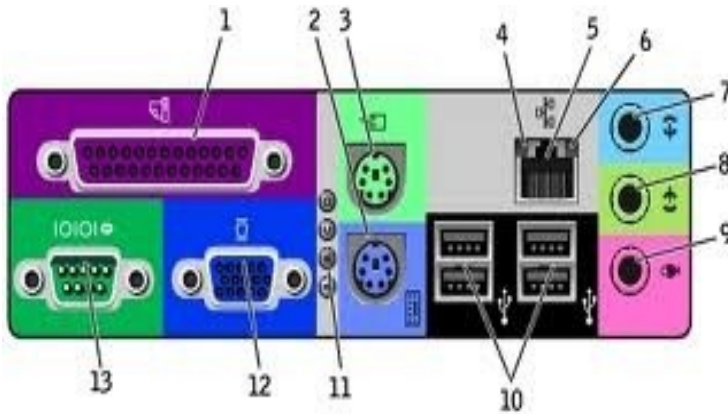


Here data is converted to signals (electromagnetic) at sender side and vice versa at receiver side.

Functions of physical layer

Defines the electrical and physical specifications for devices (interface and Tx medium).

This includes the layout of pins, voltages, cable specifications, Encoding (NRZ,RZ etc), modulation data rate, simplex/duplex, point to point/ multipoint, topology(bus/star etc.)



- **Devices working at PL:**

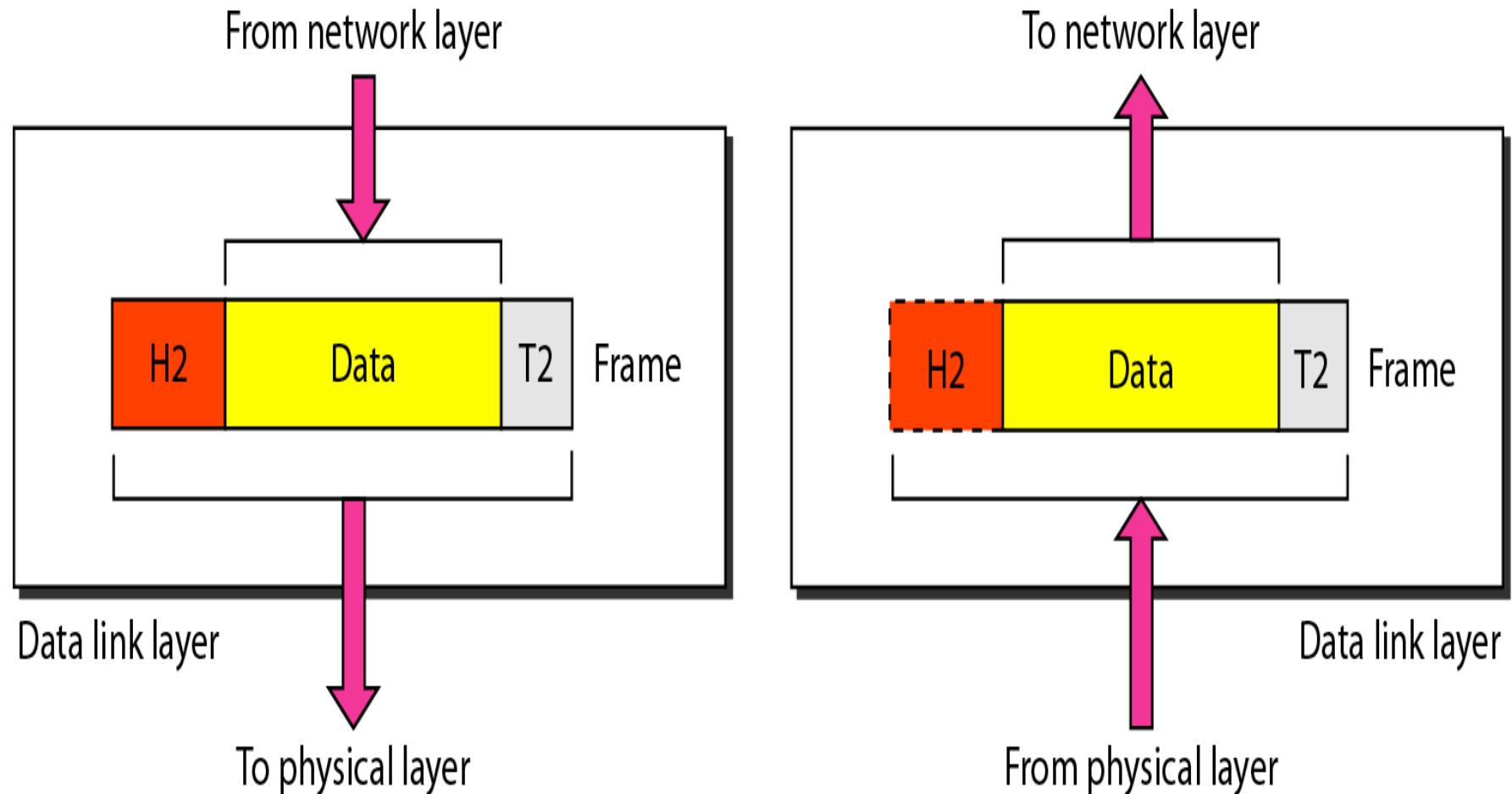
NIC, Hubs, repeaters,

- **Examples of standards defined for PL**

Token ring, FDDI, ITU-T G.hn and IEEE 802.11, as well as personal area networks such as Bluetooth and IEEE 802.15.4.

Data link layer

The data link layer is responsible for moving frames from one node to the next.



Functions of DLL:

- **Framing** : converts bits received from NL into frames
- **Physical addressing**: in case of multipoint , receiver's address is added by sender in frame
- **Flow control**: if Rx speed is less than TX , Tx is told to slow down
- **Error control** : by adding error detection & correction (for damaged, duplicate or lost frames)

in case of lost or damaged frames :Tx retransmits frames in case of lost or damaged or Rx corrects damaged frame at its own end.

in case of duplicate frames: Rx discards them, keeps only one.

- **Access control**: in case of common link between many devices, link can be given to only one device at a time

DLL is split in two sub layers : LLC and MAC

protocols which have functions of DLL

ARCnet

ATM

Cisco Discovery Protocol (CDP)

Controller Area Network (CAN)

Econet

Ethernet

Ethernet Automatic Protection Switching (EAPS)

Fiber Distributed Data Interface (FDDI)

Frame Relay

High-Level Data Link Control (HDLC)

IEEE 802.2

IEEE 802.11 wireless LAN

Link Access Procedures, D channel (LAPD)

LocalTalk

Multiprotocol Label Switching (MPLS)

Point-to-Point Protocol (PPP)

Serial Line Internet Protocol (SLIP) (obsolete)

Spanning tree protocol

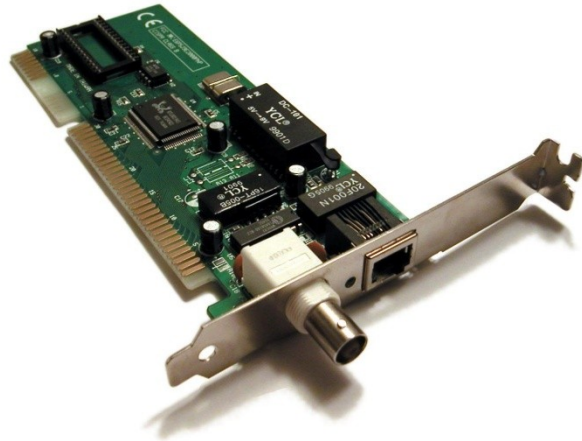
StarLan

Token ring

and most forms of serial communication.

Devices working at DLL

NIC

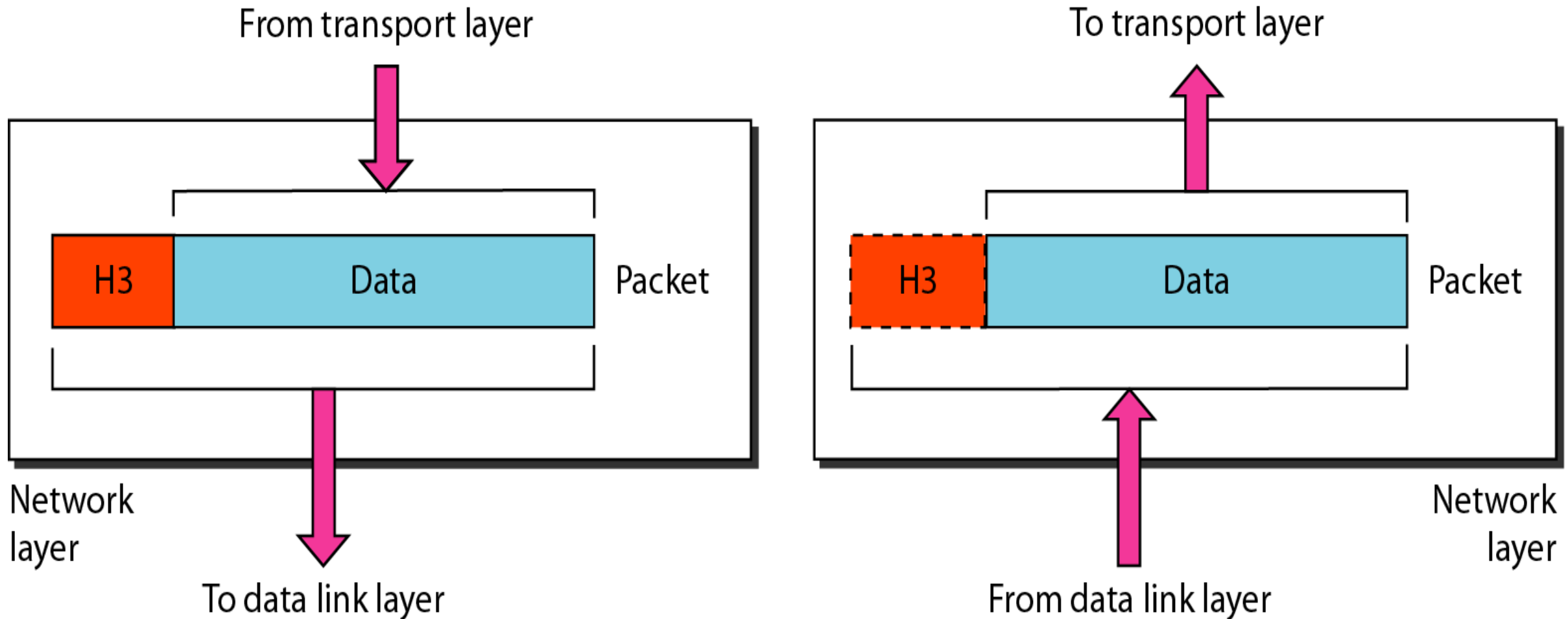


switch



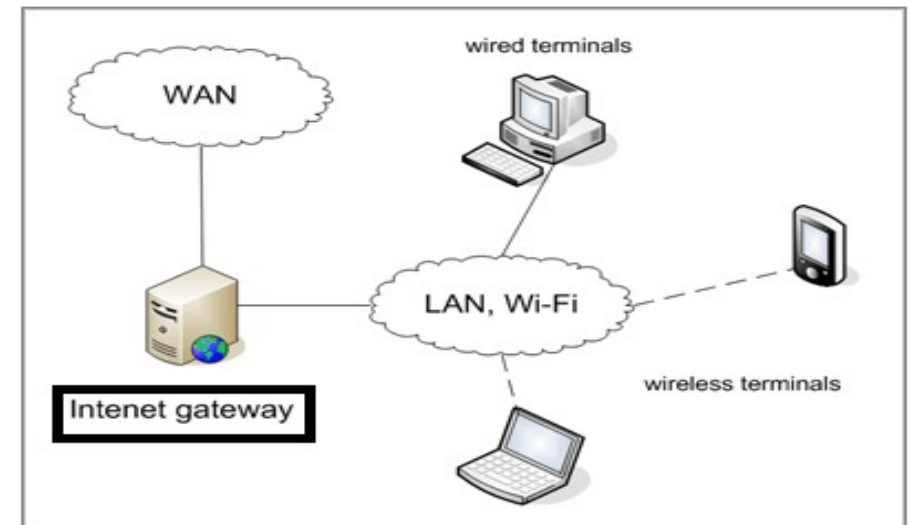
Network layer

The network layer is responsible for the delivery of individual packets from the source host(node) to the destination host(node) connected in different networks



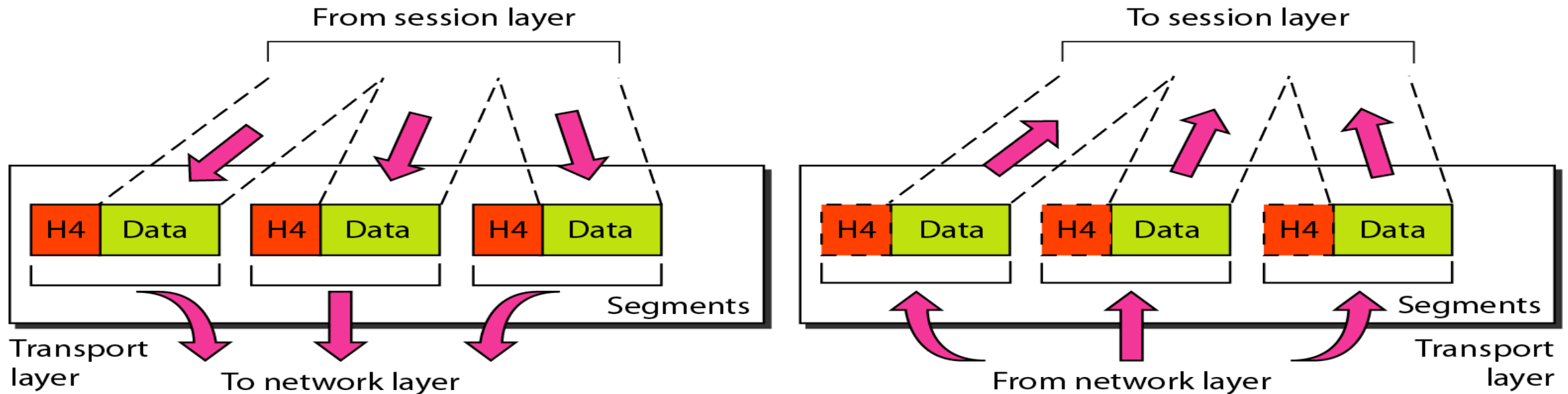
Functions of NL:

- **Logical addressing:** (DLL uses physical address, but across different networks a logical address is required) which is called Internet Protocol (IP) address.
- **Routing:** protocols to decide to which path a packet has to be sent so that it reaches RX.
- **Connection model:** connection-oriented and connectionless communication
- **Devices having NL :-Router, Gateway**

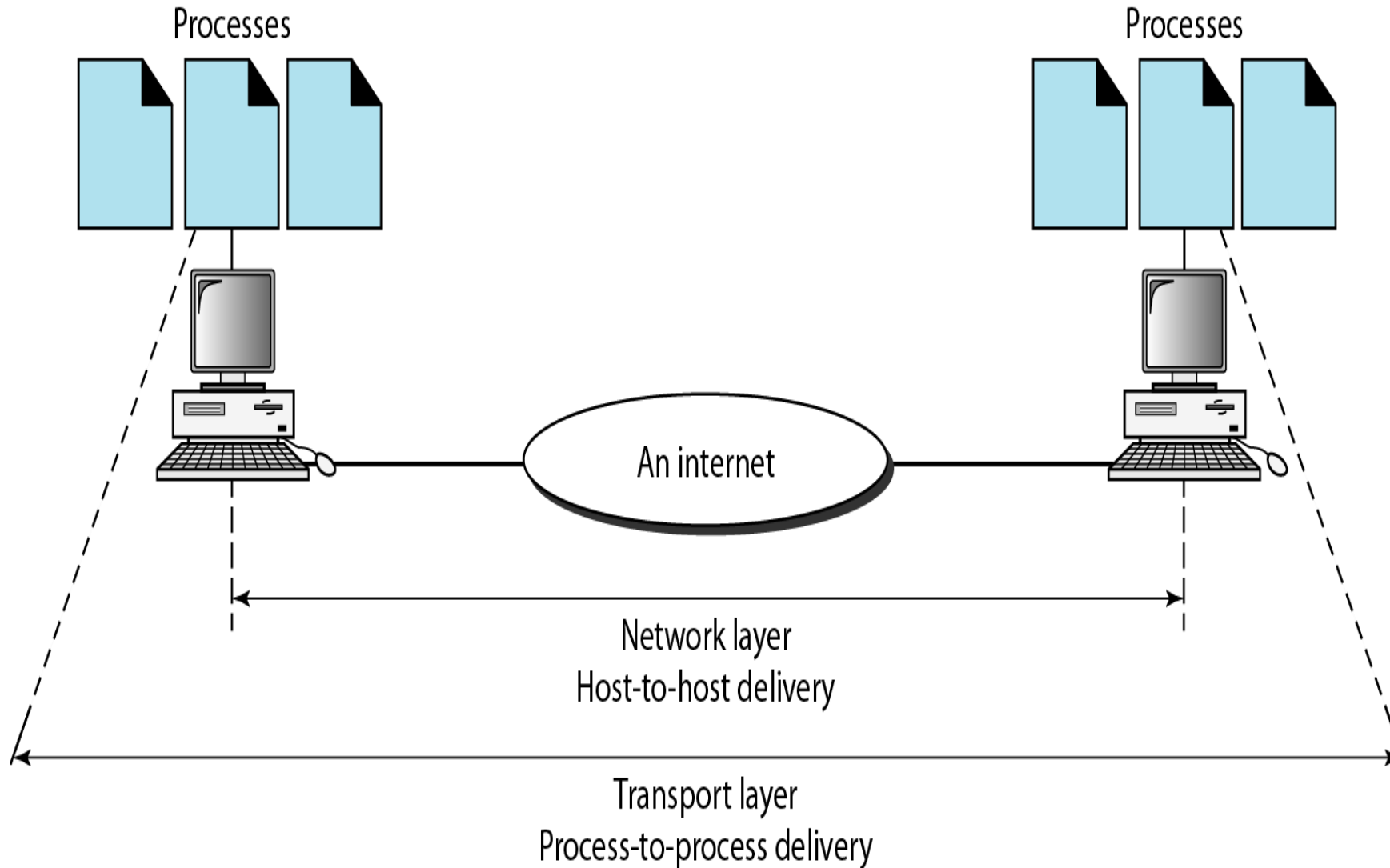


Transport layer

- The transport layer is responsible for the delivery of **entire message** from one **process/application** to another.
- A process is an application program running on a host. In destination there can be no. of applications running . For this each process must be given an address called **port address**.
- NL does not recognize any relationship between packets of same message , but TL can . So TL at Rx combines all packets of same message and gives it to intended application/process



Reliable process-to-process delivery of a message



Functions of Transport layer

- **Segmentation and reassembly:** Segmentation of message at TX and reassembly at Rx.
- **Connection control :** connection oriented or connectionless.
- **Flow control :** similar to DLL but here it is end to end. In DLL it is hop to hop (node to node)
- **Error control:** similar to DLL but here it is process to process than hop to hop as in DLL.
To make sure entire message arrives at Rx w/o error(loss, duplication, damage)

Eg of protocols used at TL

- Transmission Control Protocol (TCP)
- User Datagram Protocol (UDP)

The Session Layer

Functions (services) of session layer

- **controls the dialogues** (connections) between computers.
- It **establishes, manages and terminates** the connections between the local and remote application.
- **Authentication**(user name , password)
- **full-duplex, half-duplex, or simplex** operation,
- **Permissions** (read only , modify , delete)
- **Session restoration (checkpointing and recovery)**

It allows information of different streams, perhaps originating from different sources, to be properly combined or synchronized.(eg IDM)

Session Layer protocols

- ADSP, AppleTalk Data Stream Protocol
- ASP, AppleTalk Session Protocol
- H.245, Call Control Protocol for Multimedia Communication
- ISO-SP, OSI Session Layer Protocol (X.225, ISO 8327)
- iSNS, Internet Storage Name Service
- L2F, Layer 2 Forwarding Protocol
- L2TP, Layer 2 Tunneling Protocol
- NetBIOS, Network Basic Input Output System
- **PAP, Password Authentication Protocol**
- PPTP, Point-to-Point Tunneling Protocol
- **RPC, Remote Procedure Call Protocol**
- RTCP, Real-time Transport Control Protocol
- **SMPP, Short Message Peer-to-Peer**
- SCP, Secure Copy Protocol
- **SSH, Secure Shell**
- ZIP, Zone Information Protocol
- SDP, Sockets Direct Protocol

Presentation Layer services

- **Translation** :for different syntax and semantics. It takes care of syntactical differences in data representation within the end systems (source and destination host). Eg.
 - 1) the conversion of an EBCDIC-coded text file to an ASCII-coded file
 - 2) issues of string representation - whether they use the Pascal method (an integer length field followed by the specified amount of bytes) or the C/C++ method (null-terminated strings, i.e. "this is a string\0").
- **Encryption** : for privacy
- **Compression** : for data transfer efficiency

Presentation Layer Protocol Examples

- AFP, Apple Filing Protocol
- **ASCII**, American Standard Code for Information Interchange
- **EBCDIC**, Extended Binary Coded Decimal Interchange Code
- ICA, Independent Computing Architecture, the Citrix system core protocol
- LPP, Lightweight Presentation Protocol
- NCP, NetWare Core Protocol
- NDR, Network Data Representation
- RDP, Remote Desktop Protocol
- XDR, eXternal Data Representation
- **X.25 PAD**, Packet Assembler/Disassembler Protocol

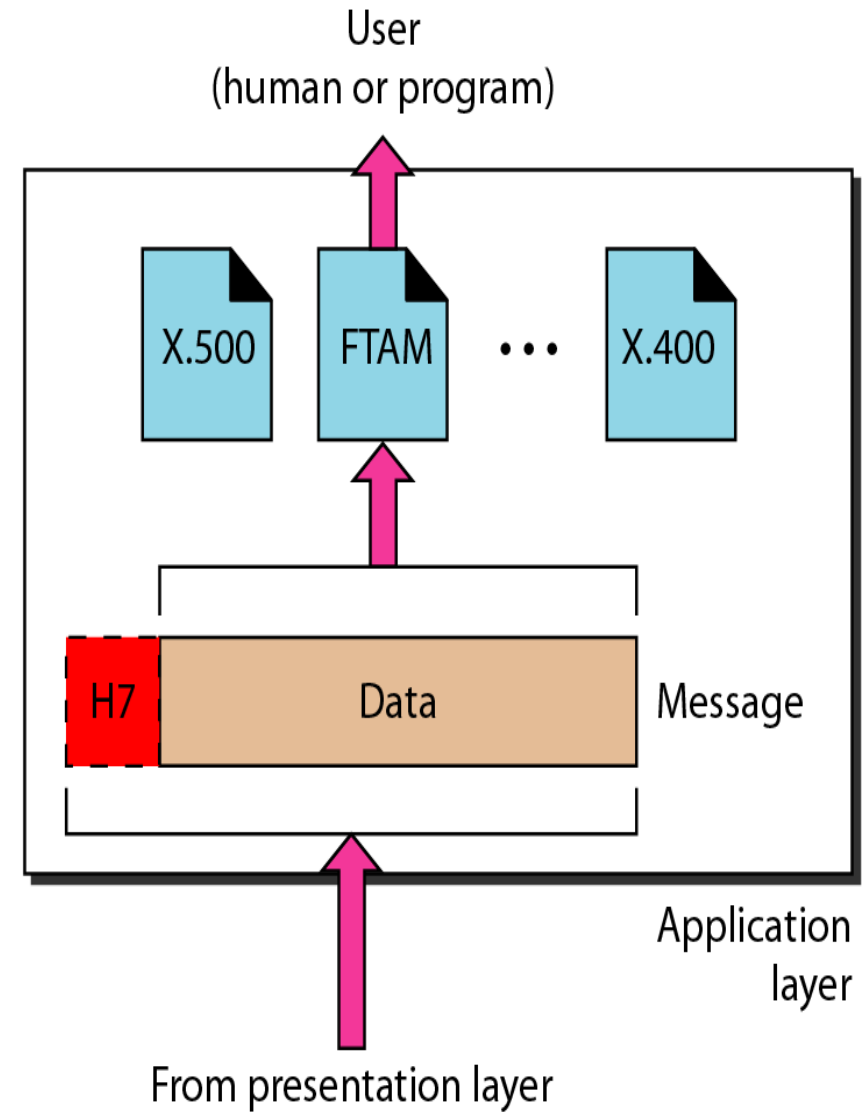
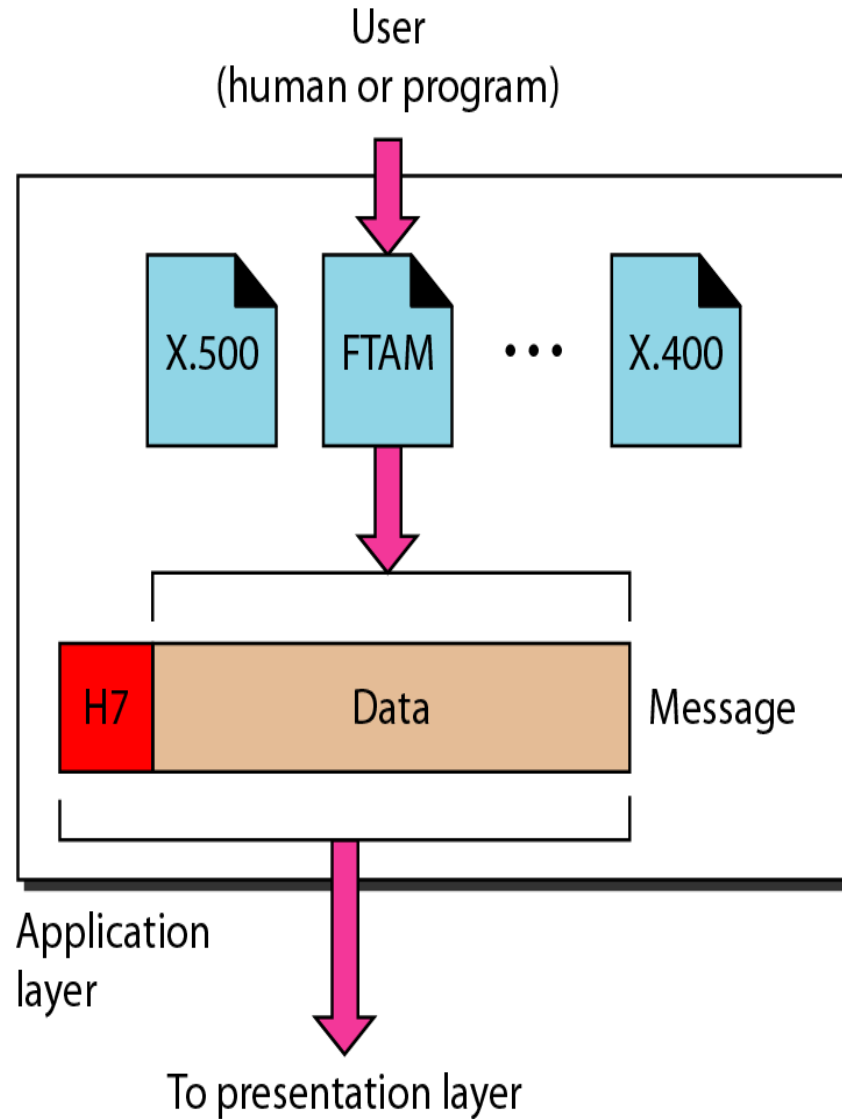
Application Layer

- This layer interacts with software applications that implement a communicating component.

Such as email, remote file access & transfer, shared database mgmt etc.

Eg when using MS word if you need online help , or to open link embedded in document , its communication component takes over.

Application layer



Application layer protocols

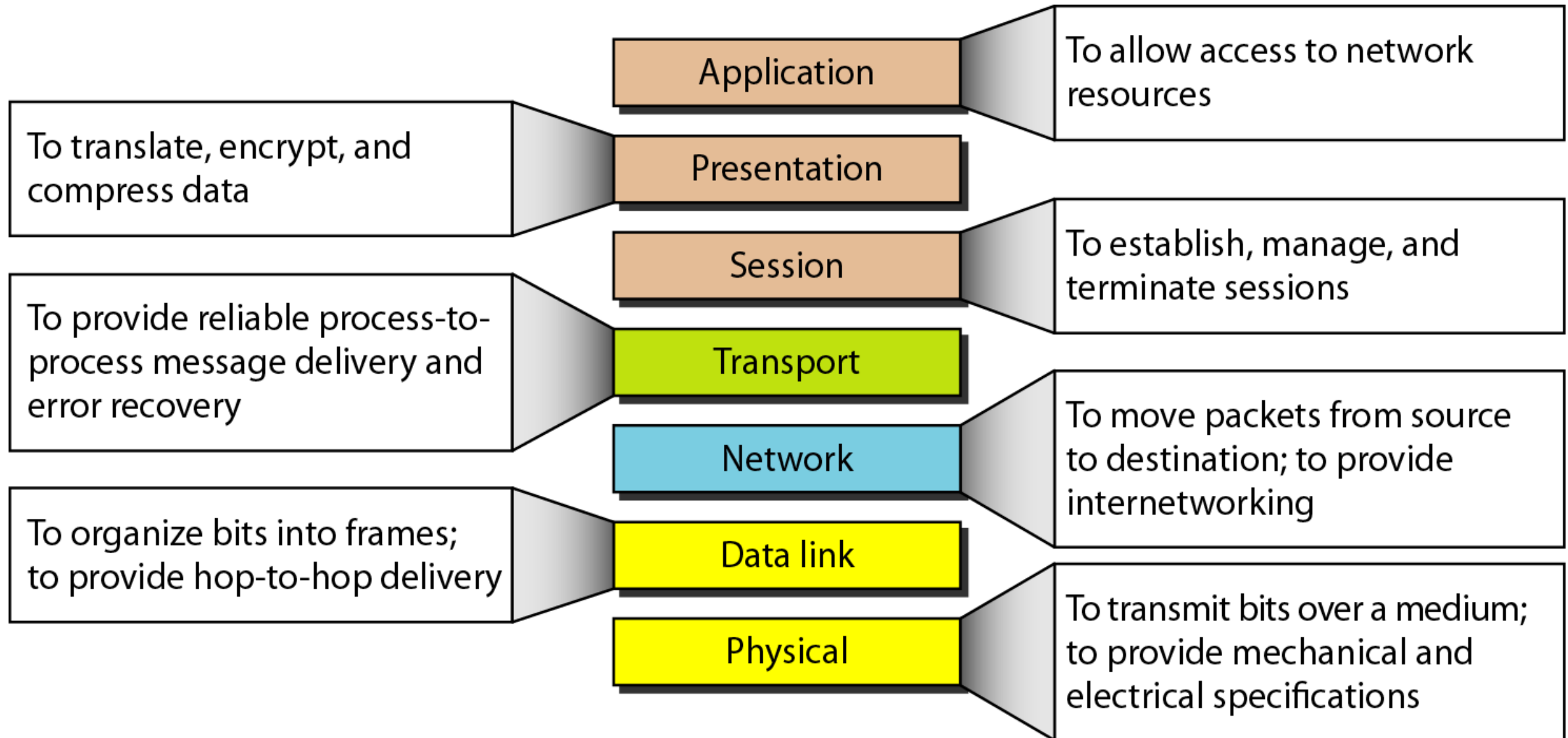
- 9P, Plan 9 from Bell Labs distributed file system protocol
- AFP, Apple Filing Protocol
- APPC, Advanced Program-to-Program Communication
- AMQP, Advanced Message Queuing Protocol
- BitTorrent
- Atom Publishing Protocol
- BOOTP, Bootstrap Protocol
- CFDP, Coherent File Distribution Protocol
- DDS, Data Distribution Service
- **DHCP, Dynamic Host Configuration Protocol**
- DeviceNet
- **DNS, Domain Name System (Service) Protocol**
- eDonkey
- ENRP, Endpoint Handlespace Redundancy Protocol

- TCAP, Transaction Capabilities Application Part
- TDS, Tabular Data Stream
- **TELNET, Terminal Emulation Protocol of TCP/IP**
- TFTP, Trivial File Transfer Protocol
- TSP, Time Stamp Protocol
- VTP, Virtual Terminal Protocol
- Waka, an HTTP replacement protocol
- Whois (and RWhois), Remote Directory Access Protocol
- WebDAV
- **X.400, Message Handling Service Protocol(for e mails)**
- **X.500, Directory Access Protocol (DAP) (to support x.400)**
- XMPP, Extensible Messaging and Presence Protocol

Summary

Layer	Functions	Address , data unit	Device	Examples
A	Allow access to N/W ,Helps Application that need to communicate to other nodes eg. email, remote file access & transfer, website access	URL , APDU	computer	HTTP, FTP, DNS, X.400
P	Translation , Encryption , Compression	--	computer	telnet, X.25 PAD,
S	Session mgmt (start, manage, stop) ,Authentication , Permissions Session restoration	-- , message	computer	PAP,RPC ,SMPP
T	Process to process delivery , Segmentation and reassembly , Connection control , Flow control , Error control	Port address, (16 bit), Message, segment	computer	TCP, UDP
N	Source to destination delivery , Routing , Connection model	Logical (IP) address (32 bit) , Packet	Router, gateway	IP , ICMP, IGMP,
D	Node to Node delivery , Framing , Flow control , Error control , Access control	Physical (MAC) address, (48 bit), Frame	NIC, switch	802.11, 802.2 , 802. (eternet) ,PPP
P	Convert data to signal and vice versa , electrical and mechanical specifications of Tx media , encoding, mux ,data rate	Bit	NIC, hub ,repeater	802.11,RS232, SONET, 10baseT, X.25

Summary of functions of layers



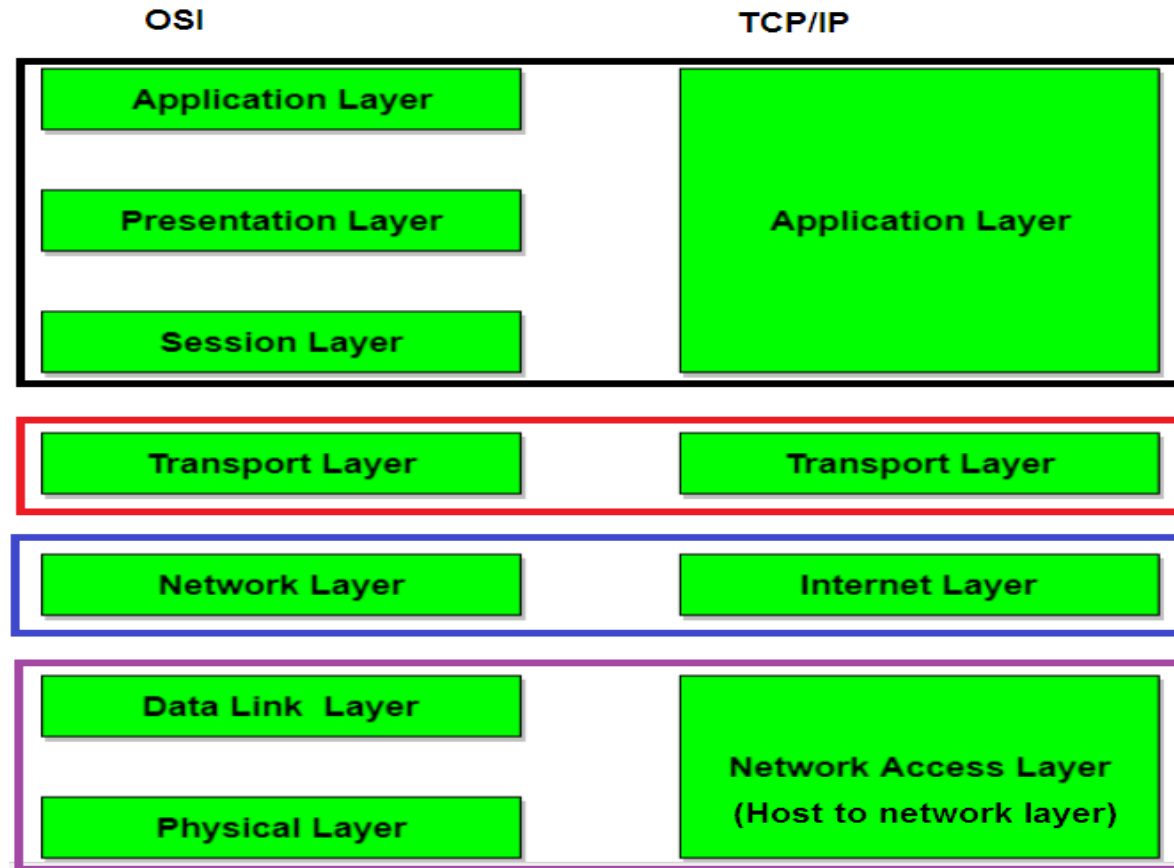
Internet Protocol Suite (TCP/IP) PROTOCOL SUITE

The **Internet Protocol Suite** (commonly known as **TCP/IP**) is the **set of** communications protocols used for the Internet and other similar networks.

It is named from two of the most important protocols in it: the Transmission Control Protocol (**TCP**) and the Internet Protocol (**IP**), which were the first two networking protocols defined in this standard.

Apart from these two , there are various other protocols in Internet protocol suite.

OSI and TCP/IP mapping



The TCP/IP protocol suite was developed prior to the OSI model. Therefore, the layers in the TCP/IP protocol suite do not exactly match those in the OSI model

TCP/IP does not define any specific protocol at PL and DLL , it supports all Protocols.

Internet layer

Internet layer Uses internetworking protocol (IP)

It also has protocols along with IP : ICMP,IGMP,ARP,RARP.

IP handles source to destination system delivery

IP is :

Connection less

Unreliable

- **Connection less** , goes out automatically without determining whether the receiver is ready, or even whether a receiver exists. (advantage over connection-oriented is that ,it has low overhead. It also allows multicast and broadcast. The packets are sent from one party to another with no need for connection establishment or connection release. The packets are not numbered; they may be delayed or lost or may arrive out of sequence..
- **connection oriented** where a communication session or a semi-permanent **connection** is established between the sender and the receiver before data transfer, and where a stream of data is delivered in the same order as it was sent.
- **Unreliable** : No error checking whether packet has reached the RX without any error. IP packets are called datagrams which can travel along different paths. So there may be no order when received by Rx or may get lost or duplicated.

- **Address Resolution Protocol (ARP)** is the method for finding a host's link layer (physical/MAC) address when only its Internet Layer (logical/IP) address is known.
- **Reverse Address Resolution Protocol (RARP)** used by a host computer to obtain its Internet Protocol (IPv4) address when it has only its physical address , but RARP is now obsolete and replaced by DHCP which has more features.
- The **Internet Control Message Protocol (ICMP)** :used by networked computers' operating systems to send error messages—indicating, for instance, that a requested service is not available or that a host or router could not be reached.

Transport layer

The protocol used in this layer is Transmission control protocol(**TCP**)

the TCP makes delivery from a process in source system to a process in destination system.

for example a Web browser and a Web server.

functions of TCP :

- provides **connection oriented** ,
- **Reliable** (error control)
- **ordered delivery** (re sequencing of segments at Rx).
- controls segment size,
- **Flow control** the rate at which data are exchanged and to reduce network traffic congestion.

The User Datagram Protocol (UDP)

- UDP uses a simple transmission model **without** hand-shaking dialogues
- So NO reliability, ordering, or data integrity. Thus, UDP provides an **unreliable** service and datagrams may arrive out of order, appear duplicated, or go missing without notice.
- UDP assumes that error checking and correction is either not necessary or performed in the application, avoiding the overhead of such processing at the network interface level.

Applications of UDP :

- the Domain Name System (DNS), streaming media applications such as IPTV, Voice over IP (VoIP), Trivial File Transfer Protocol (TFTP) and many online games.

OSI

It has 7 layers (APSTNDP)

OSI is a generic, protocol independent standard

OSI is a reference model around which the networks are built. Generally it is used as a guidance tool. More commonly used as teaching aids.

Follows vertical approach. (Corresponding Layers communicate)

OSI model has a separate Presentation layer and Session layer.

The model came first, and the protocols later

TCP/IP

It has 4 layers (ATIN)

TCP/IP model is based on standard protocols Used in internet of the world.

TCP/IP model is, in a way implementation of the OSI model.

Follows horizontal approach. (Adjacent Layers communicate)

TCP/IP does not have a separate Presentation layer or Session layer.

The protocols came first, and the model later was really just a description of the existing protocols.

Transport Layer is Connection Oriented	Transport Layer is Connection Oriented
Network Layer is both Connection Oriented and Connection less.	Network Layer is Connection less.
OSI model has a problem of fitting the protocols into the model.(some protocols combine functions of various layers) because OSI has strict boundaries;	TCP/IP does not have very strict boundaries.
OSI model defines services, interfaces and protocols very clearly and makes clear distinction between them. It is protocol independent.	In TCP/IP, services, interfaces and protocols are not clearly separated. It is also protocol dependent.
The Presentation Layer Not found in TCP/IP model In TCP/IP, this function is provided by the Application Layer.	The Session layer Not found in TCP/IP model .In TCP/IP,its characteristics are provided by the TCP protocol. (Transport Layer)

OSI MODEL

Contains 7 Layers

Uses Strict Layering resulting in vertical layers.

Supports both connectionless & connection-oriented communication in the Network layer, but only connection-oriented communication in Transport Layer

It distinguishes between Service, Interface and Protocol.

Protocols are better hidden and can be replaced relatively easily as technology changes (No transparency)

OSI reference model was devised before the corresponding protocols were designed.

TCP/IP MODEL

Contains 4 Layers

Uses Loose Layering resulting in horizontal layers.

Supports only connectionless communication in the Network layer, but both connectionless & connection-oriented communication in Transport Layer

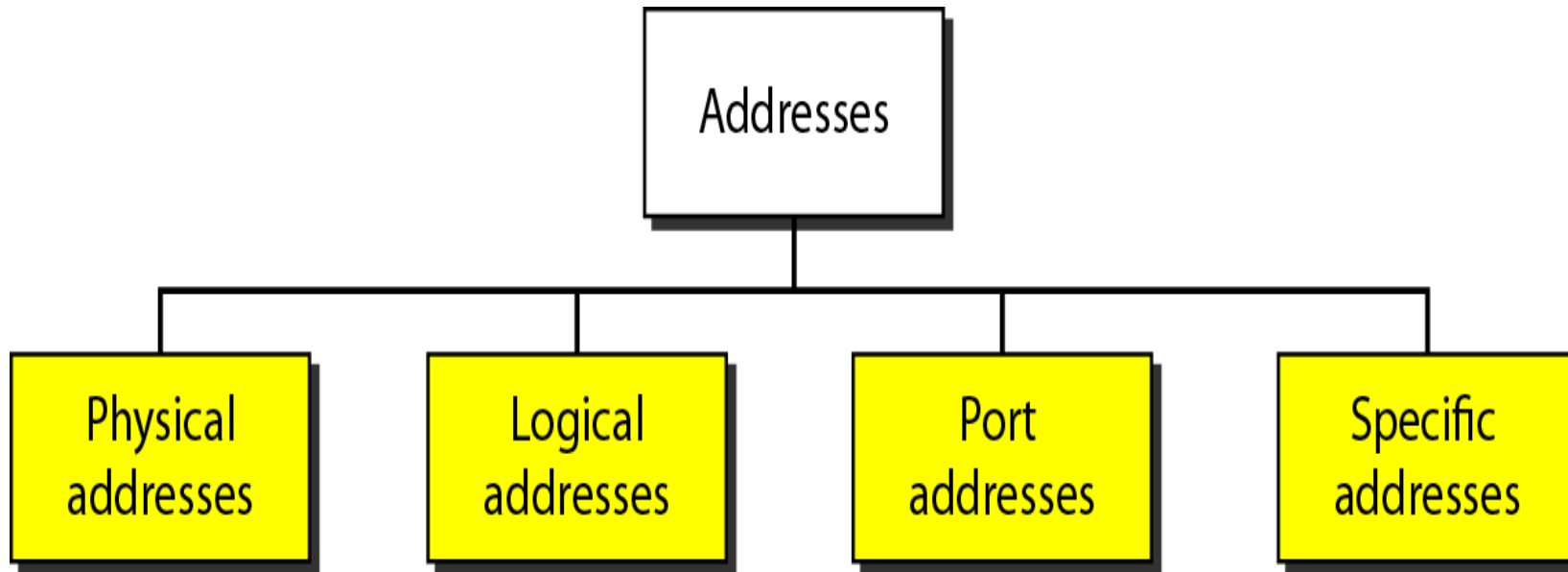
Does not clearly distinguish between Service, Interface and Protocol.

Protocols are not hidden and thus cannot be replaced easily.
(Transparency)
Replacing IP by a substantially different protocol would be virtually impossible

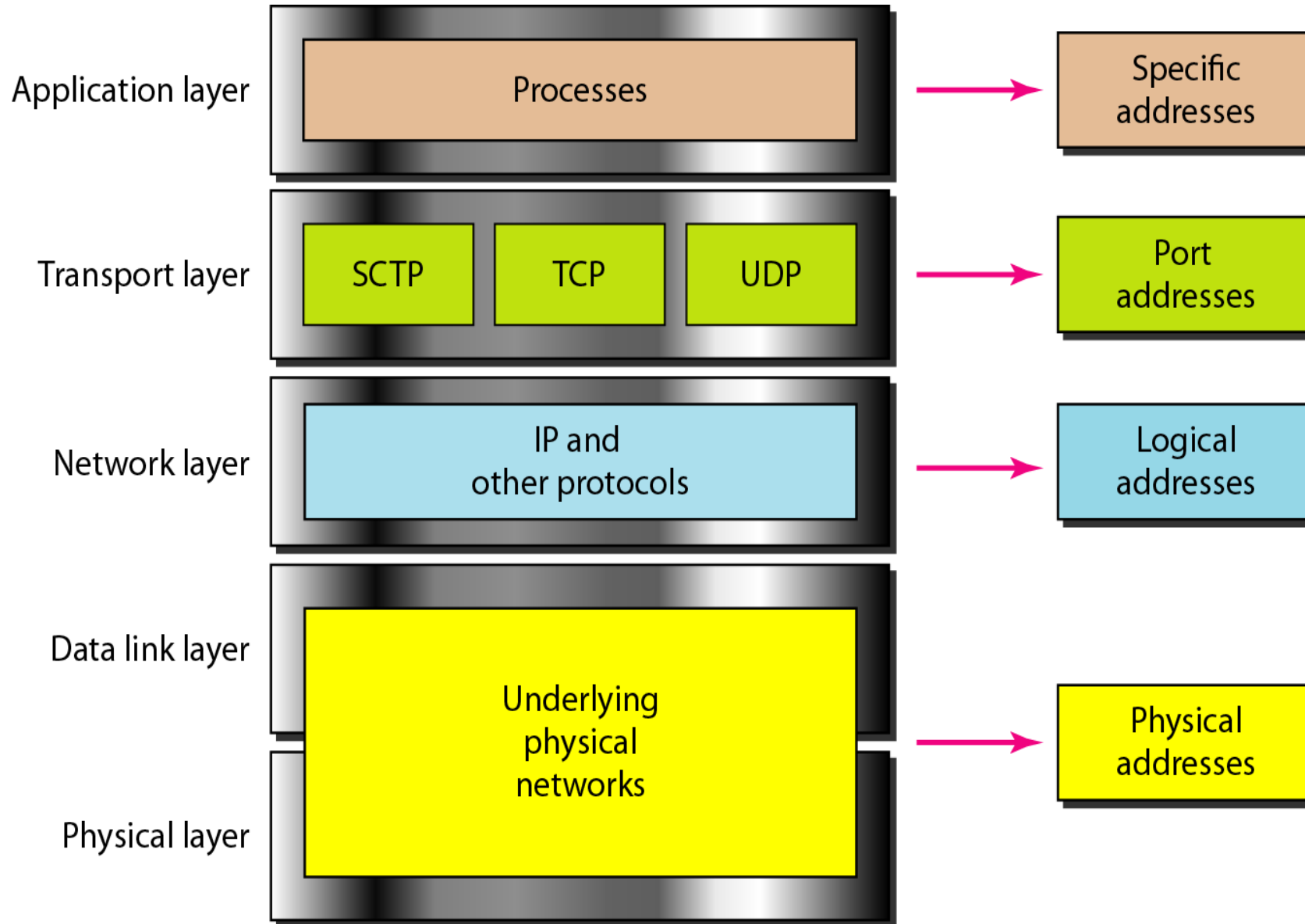
The protocols came first and the model was a description of the existing protocols

Addressing

Four levels of address are used in an internet employing the TCP/IP protocols:
physical,
logical,
port, and
specific.



Relationship of layers and address in TCP/IP

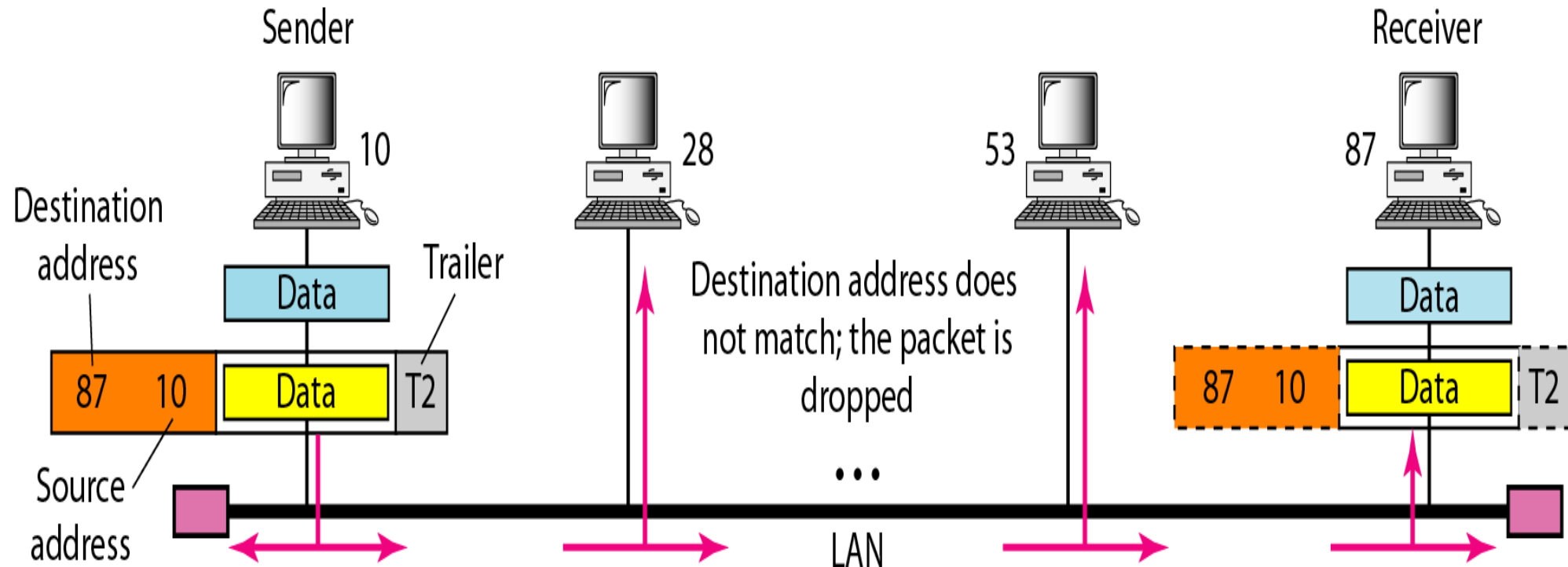


Physical address (MAC address)

In Figure a node with physical address 10 sends a frame to a node with physical address 87.

The two nodes are connected by a link (bus topology LAN).

As the figure shows, the computer with physical address 10 is the sender, and the computer with physical address 87 is the receiver.



Most local-area networks use a 48-bit (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a colon, as shown below:

A MAC address is a unique 48 bit network hardware identifier that is burned into a ROM chip on the NIC(network interface card) to identify that device on the network.

The first 24 bits is called the Organizationally Unique Identifier (OUI) and is largely manufacturer dependent. Each OUI allows for 16,777,216 Unique NIC address.

07:01:02:01:2C:4B

A 6-byte (12 hexadecimal digits) physical address.

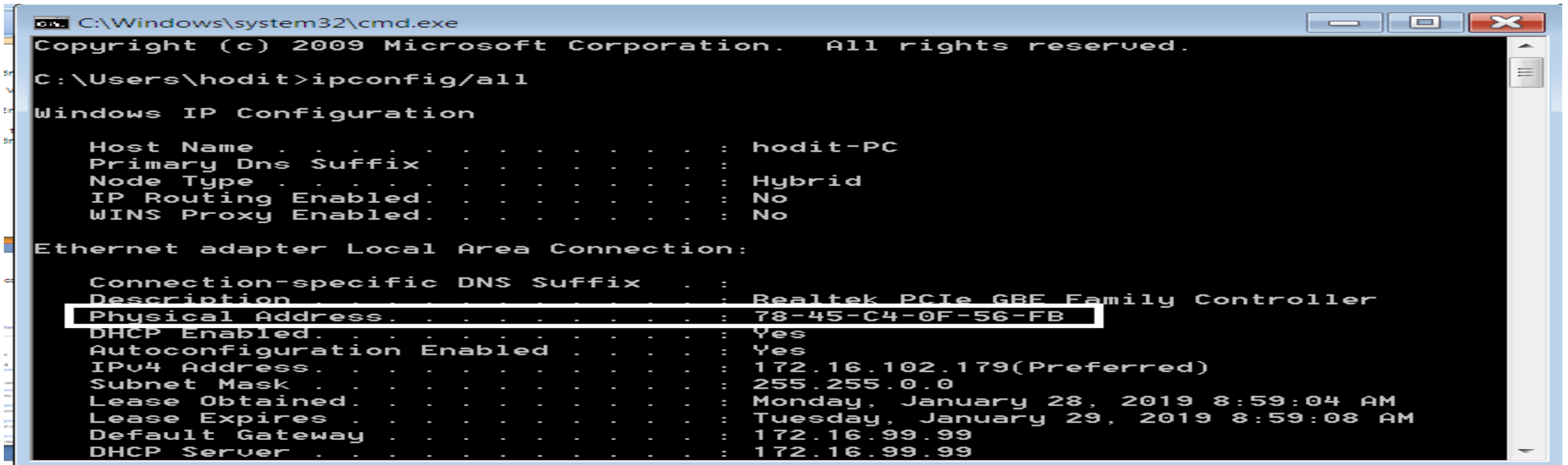
Q. How to find MAC address of computer ?

Method 1) Press the Windows Start key to open the Start screen.

Type cmd and press Enter to launch the command prompt.

Type ipconfig /all into the command prompt.

Look for the MAC address listed as Physical address



```
C:\Windows\system32\cmd.exe
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\hodit>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : hodit-PC
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . :
    Description . . . . . : Realtek PCIe GBE Family Controller
    Physical Address. . . . . : 78-45-C4-0F-56-FB
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 172.16.102.179(Preferred)
    Subnet Mask . . . . . : 255.255.0.0
    Lease Obtained. . . . . : Monday, January 28, 2019 8:59:04 AM
    Lease Expires . . . . . : Tuesday, January 29, 2019 8:59:08 AM
    Default Gateway . . . . . : 172.16.99.99
    DHCP Server . . . . . : 172.16.99.99
```

IP address

Next Figure shows a part of an internet with two routers connecting three LANs.

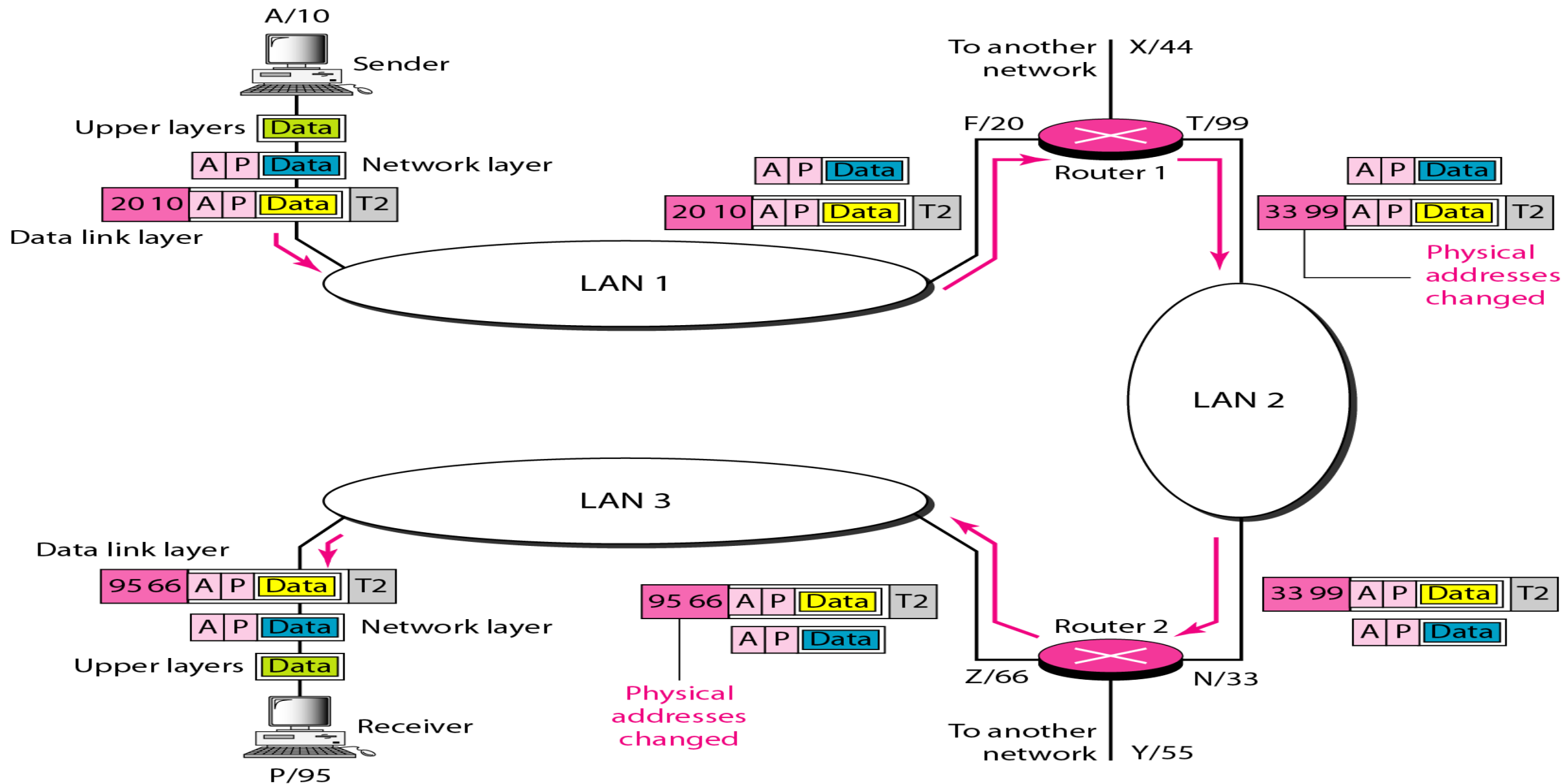
Each device (computer or router) has a pair of address (logical and physical) for each connection.

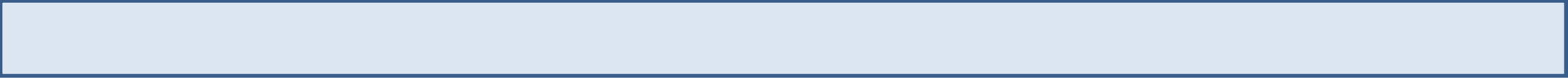
In this case, each computer is connected to only one link and therefore has only one pair of address.

Each router, however, is connected to three networks (only two are shown in the figure).

So each router has three pairs of address, one for each connection.

IP address





The physical address change from hop to hop, but the logical and port address usually remain the same.

If a computer's NIC gets faulty. It needs to be replaced with new NIC having another physical address, But you can keep the IP same

MAC works for computers in same network(ethernet) IP works for computers on different networks

A computer can find IP of a computer whose MAC is known by a protocol called ARP And vice versa by RARP

Port address

- Used at transport layer to distinguish between different processes(applications)
- It is 16 bit (so total 64000 address)
- But represented in decimal format eg 753
- Some of them are assigned to standard processes
- While remaining are available to developers.
- This port no. for application is decided by the Kernel of the OS

Well-known ports: 0–1023

[\[edit\]](#)

Port	Description	Status
0/TCP,UDP	Reserved	Official
1/TCP,UDP	TCP Port Service Multiplexer	Official
2/TCP,UDP	Management Utility	Official
3/TCP,UDP	Compression Process	Official
5/TCP,UDP	Remote Job Entry	Official
7/TCP,UDP	Echo	Official
9/TCP,UDP	Discard	Official
11/TCP,UDP	Active Users	Official
13/TCP,UDP	DAYTIME - (RFC 867 ↗)	Official
17/TCP,UDP	Quote of the Day	Official
18/TCP,UDP	Message Send Protocol	Official
19/TCP,UDP	Character Generator	Official
20/TCP	FTP - data	Official
21/TCP	FTP—control (command)	Official
22/TCP,UDP	Secure Shell (SSH)—used for secure logins, file transfers (scp , sftp) and port forwarding	Official
23/TCP	Telnet protocol—unencrypted text communications	Official
25/TCP	Simple Mail Transfer Protocol (SMTP)—used for e-mail routing between mail servers	Official
35/TCP,UDP	Any private printer server protocol	Official

Specific address

Difficult for us to remember IP address which consists of just numbers, so they are represented by user-friendly address.

Eg: `www.abc.com`

But These need to be converted back to IP for actual communication , Which is done by DNS

CHANNEL CAPACITY

- Shannon introduced the concept of *channel capacity*, the limit at which data can be transmitted through a medium.
 - The errors in the transmission medium depend on the energy of the signal, the energy of the noise, and the bandwidth of the channel.
 - Conceptually, if the bandwidth is high, we can pump more data in the channel.
 - If the signal energy is high, the effect of noise is reduced.
 - According to Shannon, the bandwidth of the channel and signal energy and noise energy are related by the formula
- $$C = W \log_2(1 + S/N)$$
- where
C is channel capacity in bits per second (bps)
 - W is bandwidth of the channel in Hz
 - S/N is the signal-to-noise power ratio (SNR).

Signal conversion methods

- **Signal Characteristics in Data Communication**
- **Signal Amplitude**
 - Amplitude of a data signal is also known as its height or magnitude. This is the most basic and inherent characteristic of any data signal employed in communications and usually represents the strength of a transmitted signal
- **Frequency**
 - Frequency is the number of cycles the data signal completes in the form of oscillation in a single second. It is another basic characteristic of a communication signal and represents the intensity of transmission and reception in data communication networks. Frequency is strictly measured in Hertz (Hz)
- **Wavelength Information**
 - Wavelength is the length of a signal, which is measured in meters (m). It is also the basic characteristic of a data signal, which is usually calculated by finding a ratio between the speed of light (c) and the frequency of the signal (f) as c/f .
- **Signal Phase**
 - The phase of a data signal is the shift of its amplitude angles, which is commonly measured in degrees. In more precise terms, this quantity represents the changes that occur within the magnitude of a data signal cycles with respect to time.

Signal conversion methods

Digital to Digital conversion

- In digital transmission data can be represented either in analog or digital form.
- The computers used the digital form to store the information.
- Therefore, the data needs to be converted in digital form so that it can be used by a computer.
- Digital to digital encoding is the representation of digital information by a digital signal.
- When binary 1s and 0s generated by the computer are translated into sequence of voltage pulses that can be propagated over a wire, this process known as digital to digital encoding

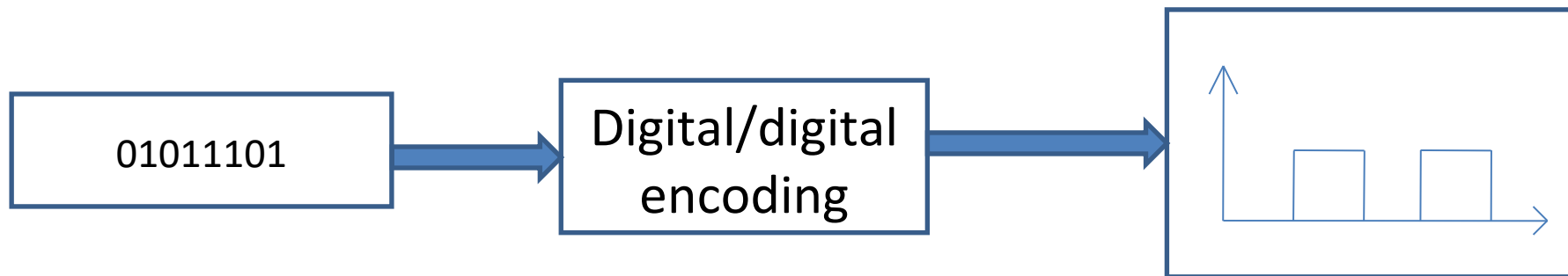


Fig) Digital to digital conversion

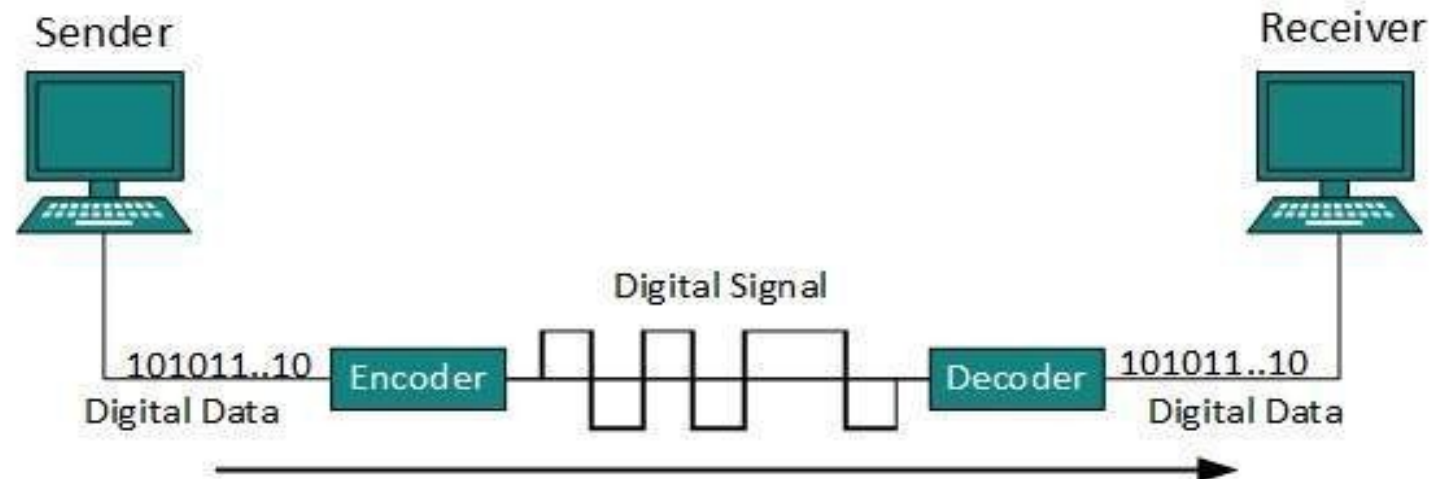
Digital to digital encoding

(Line coding) is divided into three categories: Unipolar encoding, Polar encoding, Bipolar encoding.

Line coding

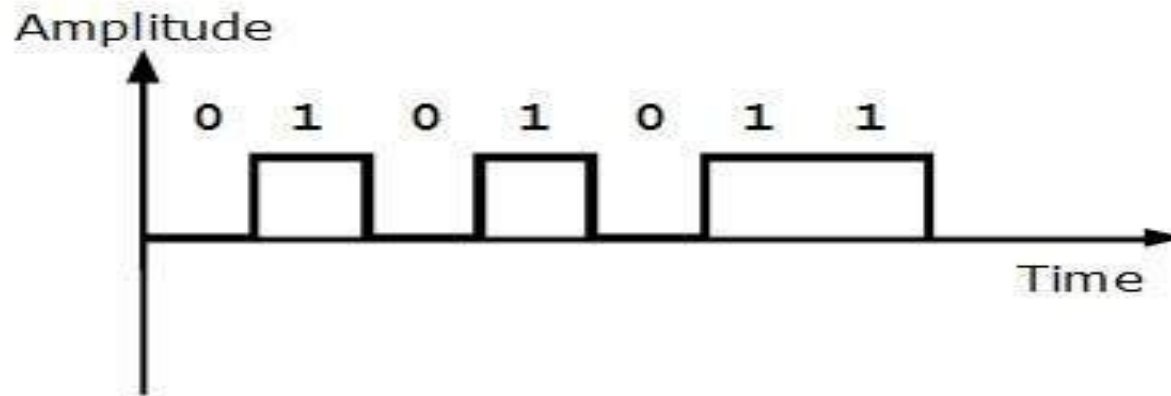
The process for converting digital data into digital signal is said to be Line Coding. Digital data is found in binary format.

It is represented (stored) internally as series of 1s and 0s. At the sender digital data are encoded into digital signal whereas at receivers end the digital data is recreated by decoding the digital signal.



Unipolar Encoding

- It is also called **Unipolar-Non-return-to-zero**, because there is no rest condition i.e. it either represents 1 or 0.
- Digital transmission system sends the voltage pulses over the medium link such as wire or cable.
- In most types of encoding, one voltage level represents 0, and another voltage level represents 1.
- The polarity of each pulse determines whether it is positive or negative.
- This type of encoding is known as Unipolar encoding as it uses only one polarity.
- In Unipolar encoding, the polarity is assigned to the 1 binary state.
- In this, 1s are represented as a positive value and 0s are represented as a zero value.
- In Unipolar Encoding, '1' is considered as a high voltage and '0' is considered as a zero voltage.
- Unipolar encoding is simpler and inexpensive to implement.



Polar NRZ

Polar

Polar encoding is an encoding scheme that uses two voltage levels: one is positive, and another is negative.

Polar NRZ

NRZ stands for Non-return zero.

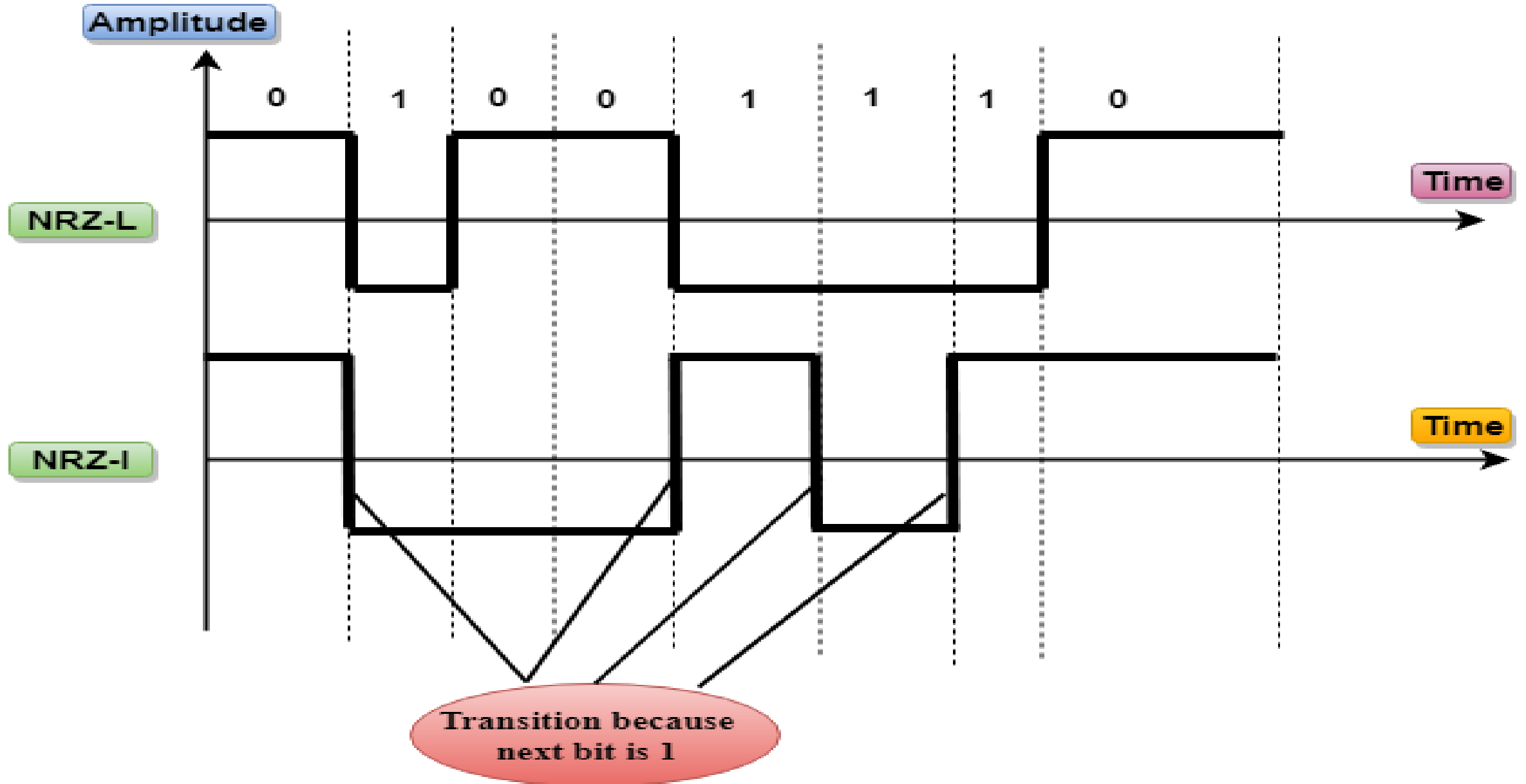
In NRZ encoding, the level of the signal can be represented either positive or negative.

The two most common methods used in NRZ are:

NRZ-L: In NRZ-L encoding, the level of the signal depends on the type of the bit that it represents. If a bit is 0 or 1, then their voltages will be positive and negative respectively. Therefore, we can say that the level of the signal is dependent on the state of the bit.

NRZ-I: NRZ-I is an inversion of the voltage level that represents 1 bit. In the NRZ-I encoding scheme, a transition occurs between the positive and negative voltage that represents 1 bit. In this scheme, 0 bit represents no change and 1 bit represents a change in voltage level.

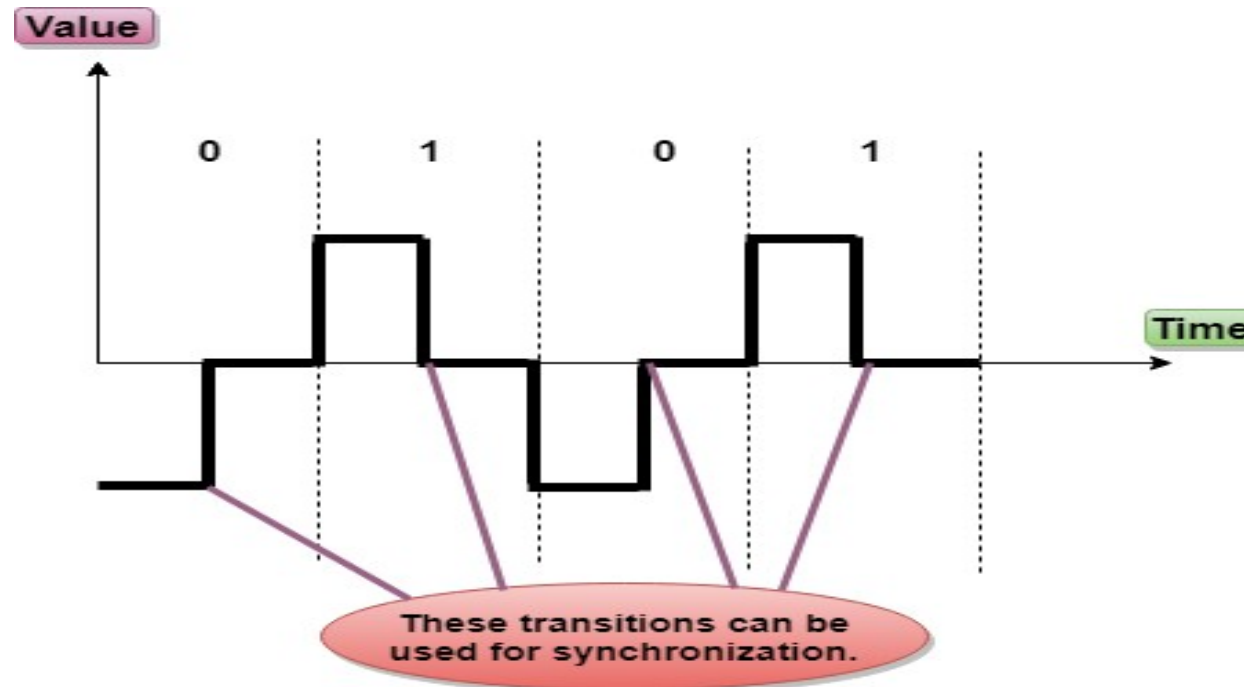
Polar NRZ



Return to zero

RZ

- RZ stands for Return to zero.
- There must be a signal change for each bit to achieve synchronization. However, to change with every bit, we need to have three values: positive, negative and zero.
- RZ is an encoding scheme that provides three values, positive voltage represents 1, the negative voltage represents 0, and zero voltage represents none.
- In the RZ scheme, halfway through each interval, the signal returns to zero.
- In RZ scheme, 1 bit is represented by positive-to-zero and 0 bit is represented by negative-to-zero.



Biphase

Biphase

Biphase is an encoding scheme in which signal changes at the middle of the bit interval but does not return to zero. Biphase encoding is implemented in two different ways:

Manchester

It changes the signal at the middle of the bit interval but does not return to zero for synchronization.

In Manchester encoding, a negative-to-positive transition represents binary 1, and positive-to-negative transition represents 0.

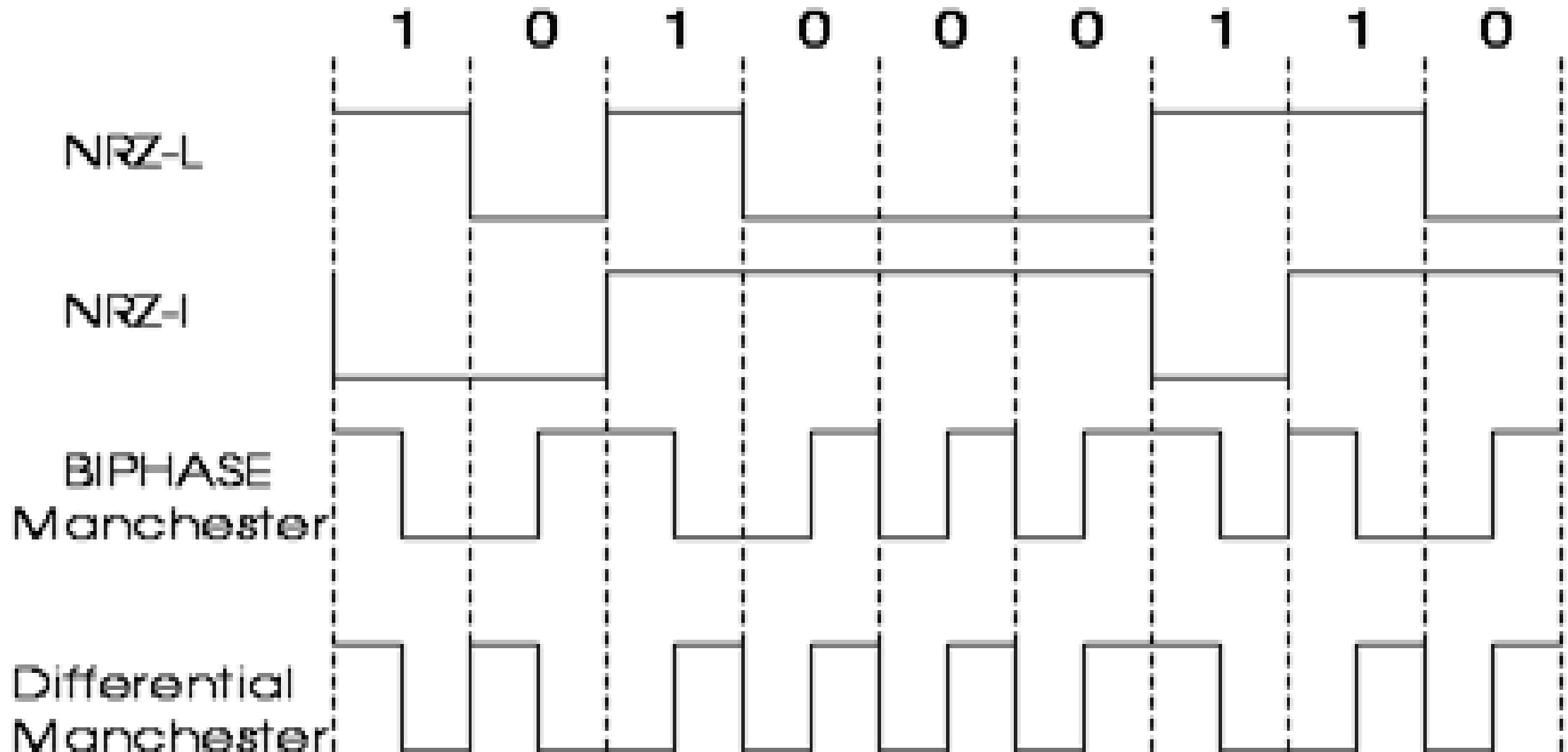
Manchester has the same level of synchronization as RZ scheme except that it has two levels of amplitude.

Differential Manchester

It changes the signal at the middle of the bit interval for synchronization, but the presence or absence of the transition at the beginning of the interval determines the bit. A transition means binary 0 and no transition means binary 1.

In Manchester Encoding scheme, two signal changes represent 0 and one signal change represent 1.

Manchester & Differential manchester

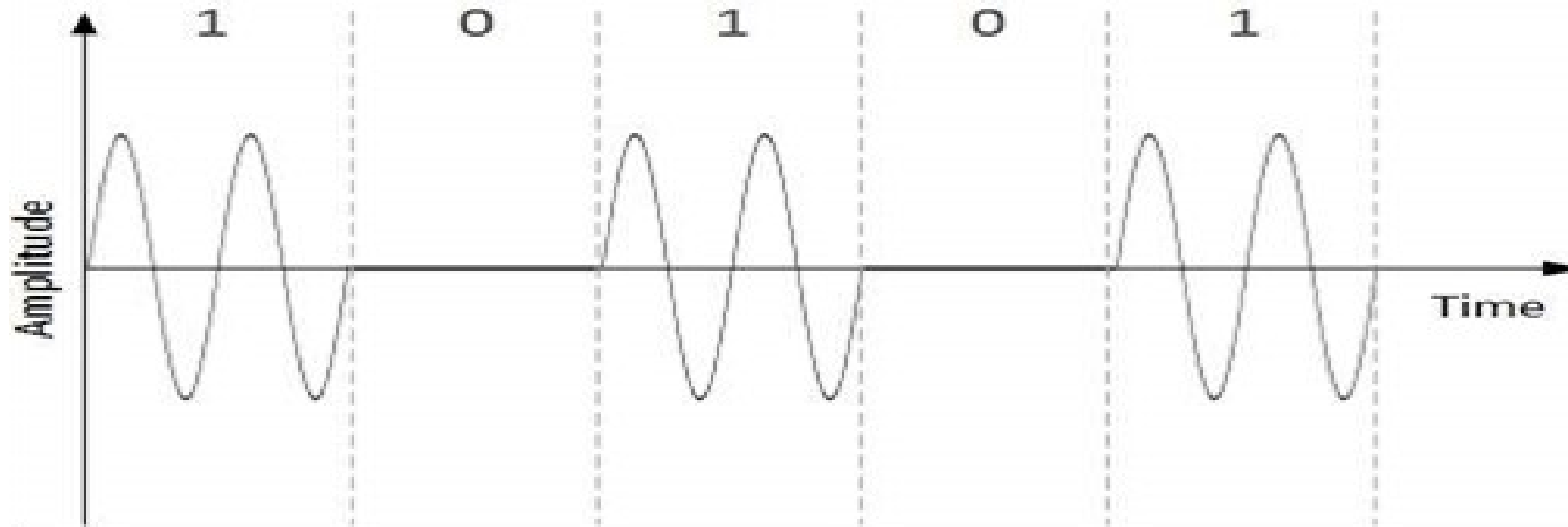


Digital Data with Analog Signals

- This method is used to send computer information over transmission channels that require analog signals, like a fiber optic networks, computer modems, cellular phone networks, and satellite systems.
- In each of this systems, an electromagnetic carrier wave is used to carry the information over great distances and connect digital information users at remote locations.
- The digital data is used to modulate one or more of the parameters of the carrier wave, This basic process is given the name "shift-keying" to differentiate it from the purely analog systems like AM and FM. As with analog modulation, there are three parameters of the carrier wave to vary and therefore three basic types of shift keying:
 - **Amplitude Shift Keying (ASK)**
 - **Frequency Shift Keying (FSK), and**
 - **Phase Shift Keying (PSK).**

Amplitude Shift Keying

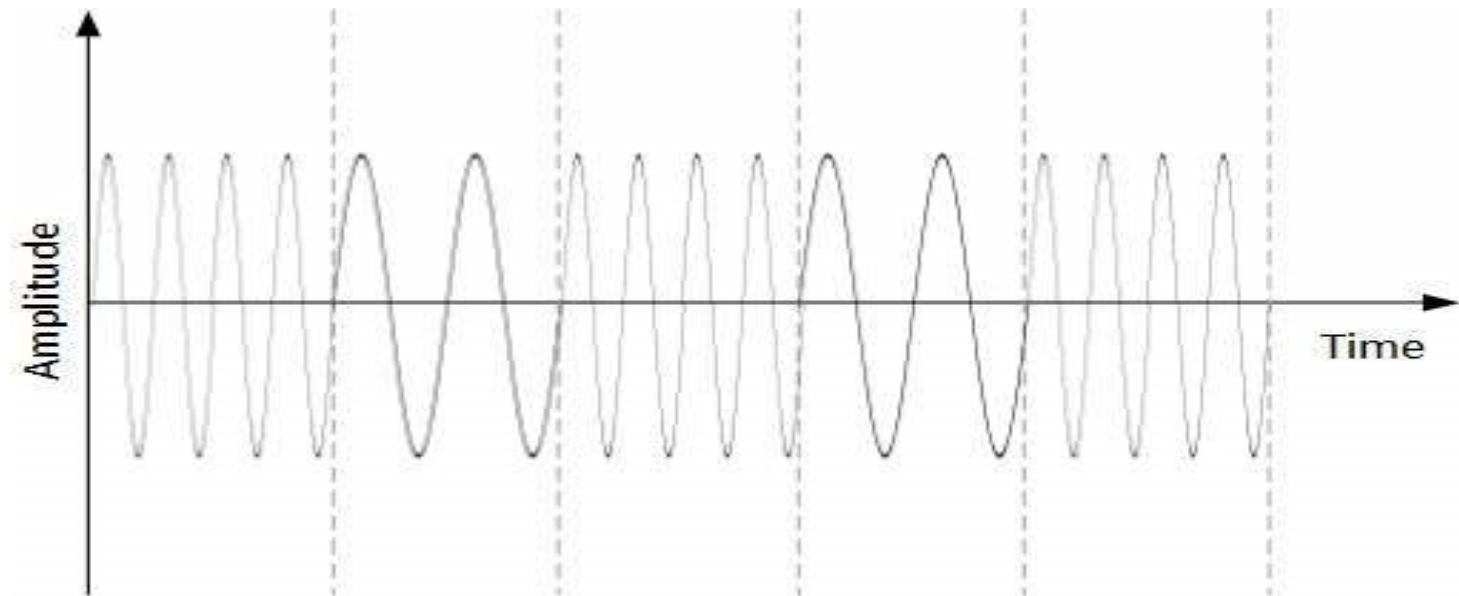
- In this conversion technique, the amplitude of analog carrier signal is modified to reflect binary data.



- When binary data represents digit 1, the amplitude is held; otherwise it is set to 0. Both frequency and phase remain same as in the original carrier signal.

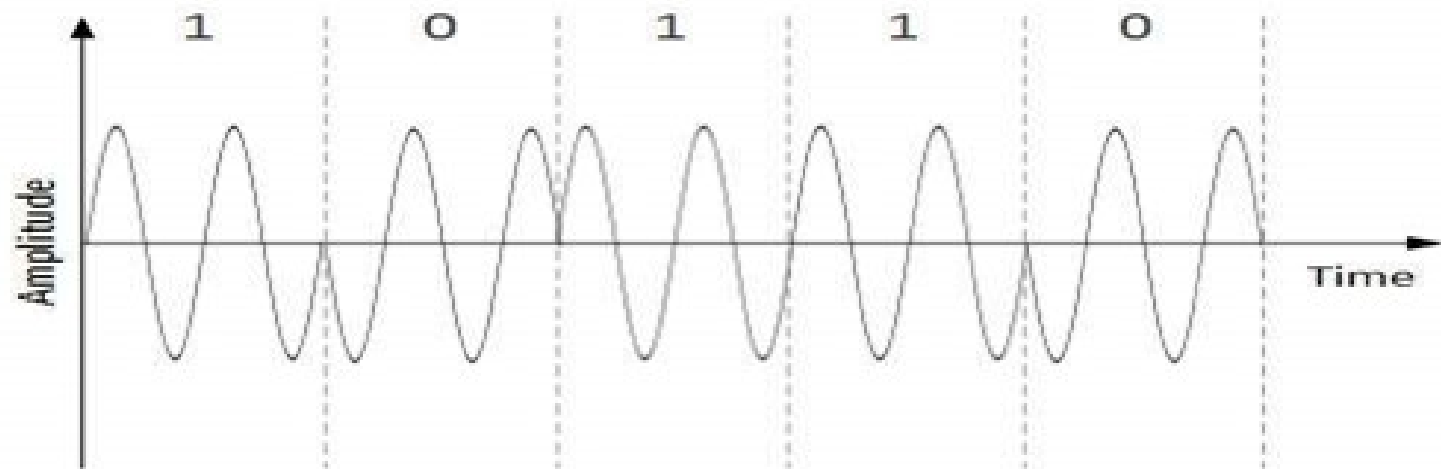
Frequency Shift Keying

- In this conversion technique, the frequency of the analog carrier signal is modified to reflect binary data.
- This technique uses two frequencies, f_1 and f_2 . One of them, for example f_1 , is chosen to represent binary digit 1 and the other one is used to represent binary digit 0. Both amplitude and phase of the carrier wave are kept intact.



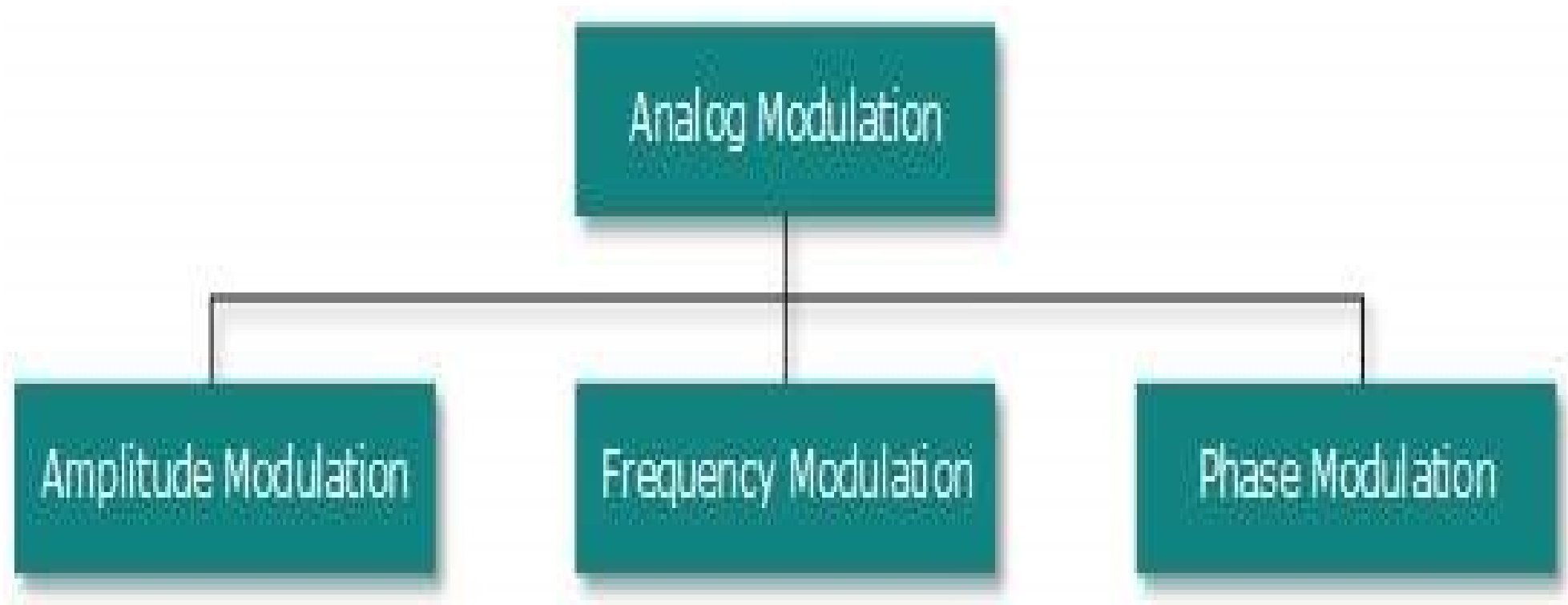
• **Phase Shift Keying**

- In this conversion scheme, the phase of the original carrier signal is altered to reflect the binary data.
- When a new binary symbol is encountered, the phase of the signal is altered. Amplitude and frequency of the original carrier signal is kept intact



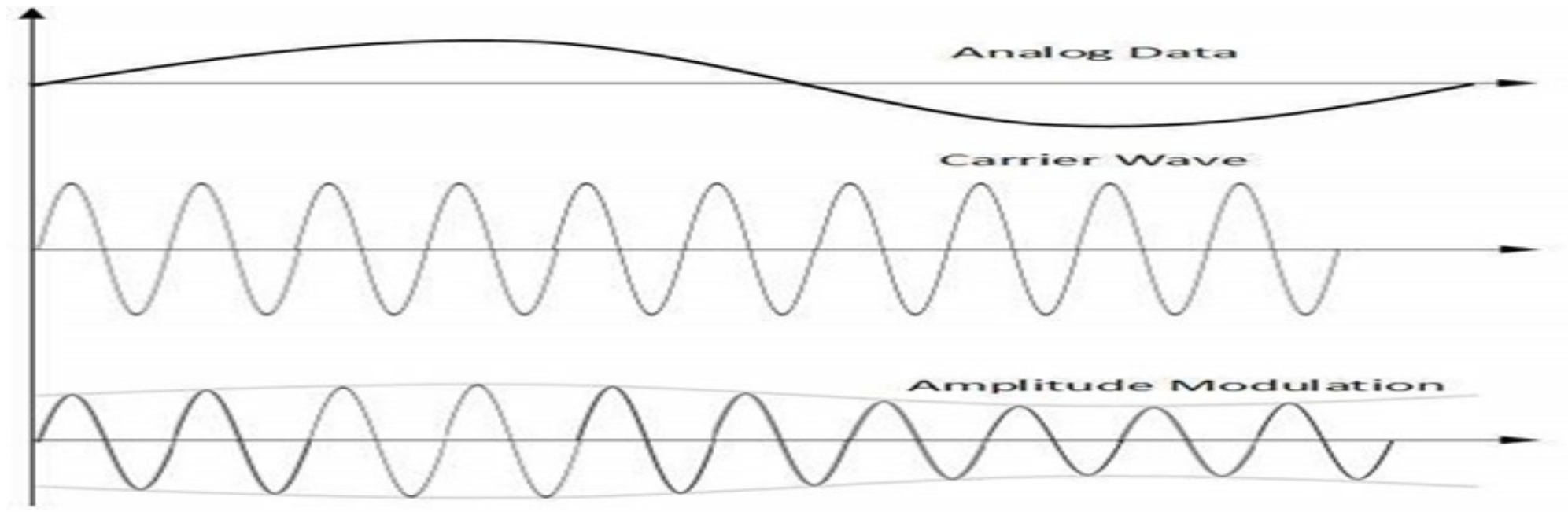
Analog-to-Analog Conversion

Analog signals are modified to represent analog data. This conversion is also known as Analog Modulation. Analog modulation is required when band pass is used. Analog to analog conversion can be done in three ways:-



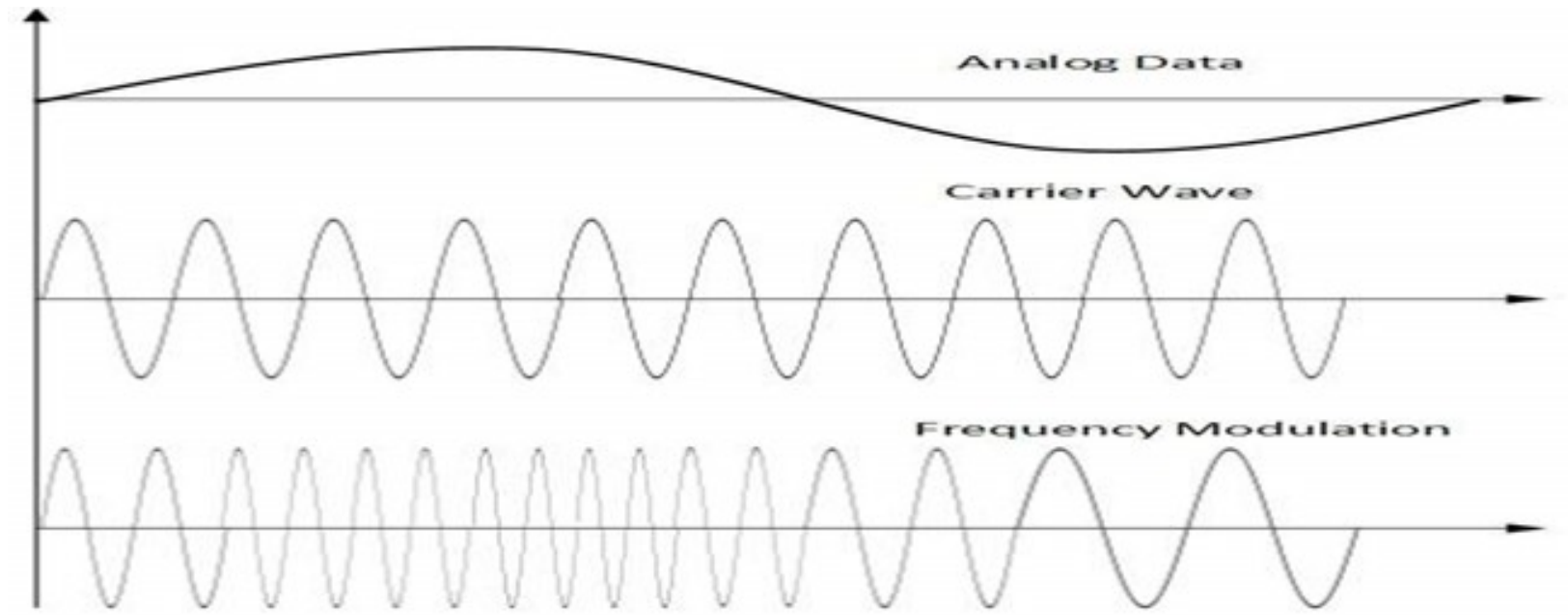
Amplitude Modulation

- In this modulation, the amplitude of the carrier signal is modified to reflect the analog data.
- Amplitude modulation is implemented by means of a multiplier. The amplitude of modulating signal (analog data) is multiplied by the amplitude of carrier frequency, which then reflects analog data. The frequency and phase of carrier signal remain unchanged.



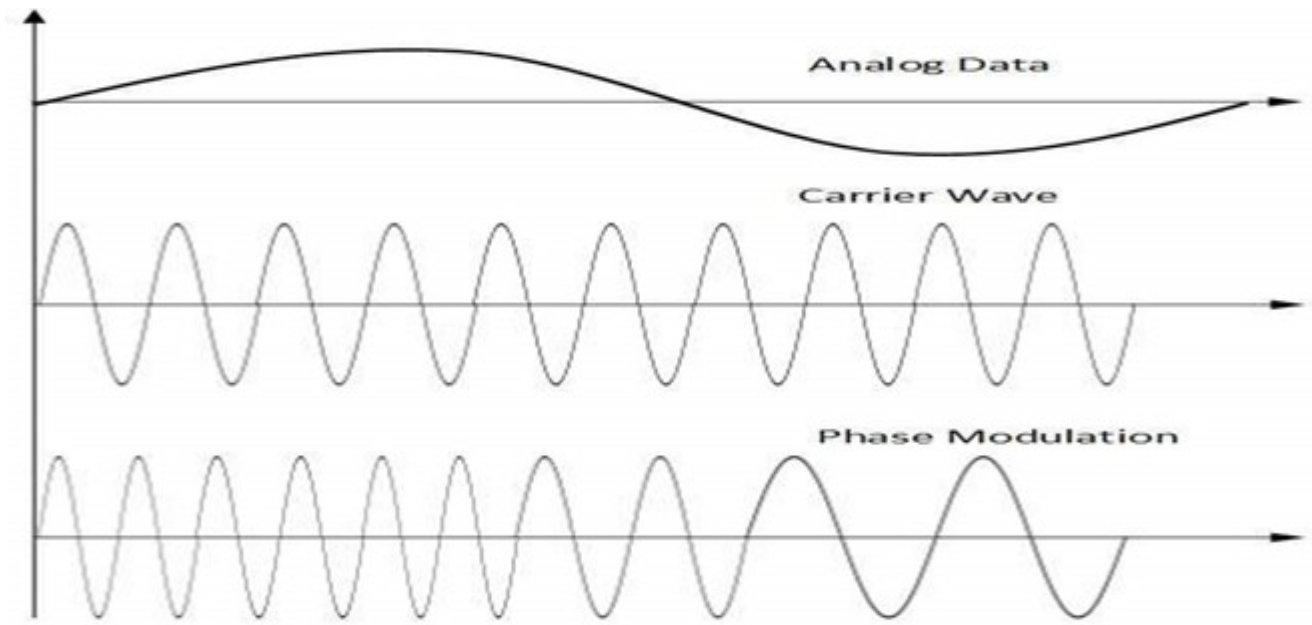
Frequency Modulation

- In this modulation technique, the frequency of the carrier signal is modified to reflect the change in the voltage levels of the modulating signal (analog data).
- The amplitude and phase of the carrier signal are not altered.



Phase Modulation

- In the modulation technique, the phase of carrier signal is modulated in order to reflect the change in voltage (amplitude) of analog data signal.
- Phase modulation is practically similar to Frequency Modulation, but in Phase modulation frequency of the carrier signal is not increased. Frequency of carrier is signal is changed (made dense and sparse) to reflect voltage change in the amplitude of modulating signal.



Nyquist Bit Rate –

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\text{BitRate} = 2 * \text{Bandwidth} * \log_2(L)$$

In the above equation, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the bit rate in bits per second.

Bandwidth is a fixed quantity, so it cannot be changed. Hence, the data rate is directly proportional to the number of signal levels.

Shannon Capacity –

In reality, we cannot have a noiseless channel; the channel is always noisy. Shannon capacity is used, to determine the theoretical highest data rate for a noisy channel:

$$\text{Capacity} = \text{bandwidth} * \log_2(1 + \text{SNR})$$

In the above equation, bandwidth is the bandwidth of the channel, SNR is the signal-to-noise ratio, and capacity is the capacity of the channel in bits per second.

Bandwidth is a fixed quantity, so it cannot be changed. Hence, the channel capacity is directly proportional to the power of the signal, as

$$\text{SNR} = (\text{Power of signal}) / (\text{power of noise}).$$

The signal-to-noise ratio (S/N) is usually expressed in decibels (dB) given by the formula:

$$10 * \log_{10}(\text{S/N})$$

so for example a signal-to-noise ratio of 1000 is commonly expressed as:

$$10 * \log_{10}(1000) = 30 \text{ dB}.$$

Network Interface Card (NIC)

In the list of the networking devices, NIC stands on the first place. Without this device, networking cannot be done.

This is also known as network adapter card, Ethernet Card and LAN card. NIC allows a networking device to communicate with the other networking device.

NIC converts the data packets between two different data transmission technologies.

A PC uses parallel data transmission technology to transmit the data between its internal parts while the media that provides connectivity between different PCs uses serial data transmission technology.

NIC



PCMCIA
Laptop NIC



1 BNC & 1 RJ-45 port
Ethernet NIC



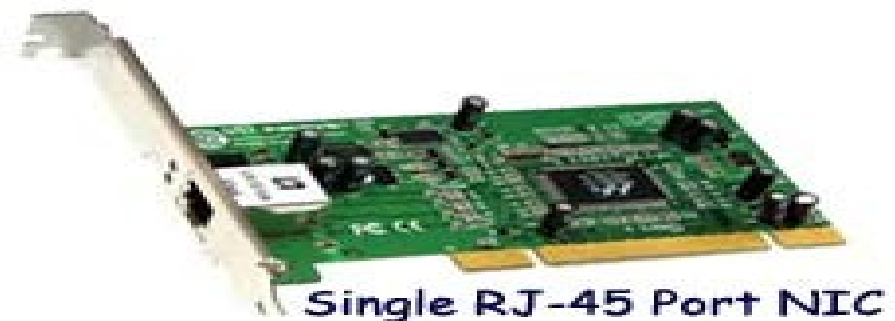
Wireless NIC



4 RJ-45 ports Ethernet NIC



Token Ring NIC



Single RJ-45 Port NIC

Hub

Hub is a centralized device that connects multiple devices in a single LAN network. When Hub receives the data signals from a connected device on any of its port, except that port, it forwards those signals to all other connected devices from the remaining ports. Usually, Hub has one or more uplink ports that are used to connect it with another Hub.

There are two types of the Hub.

Passive Hub: - It forwards data signals in the same format in which it receives them. It does not change the data signal in any manner.

Active Hub: - It also works same as the passive Hub works. But before forwarding the data signals, it amplifies them. Due to this added feature, the active Hub is also known as the repeater.

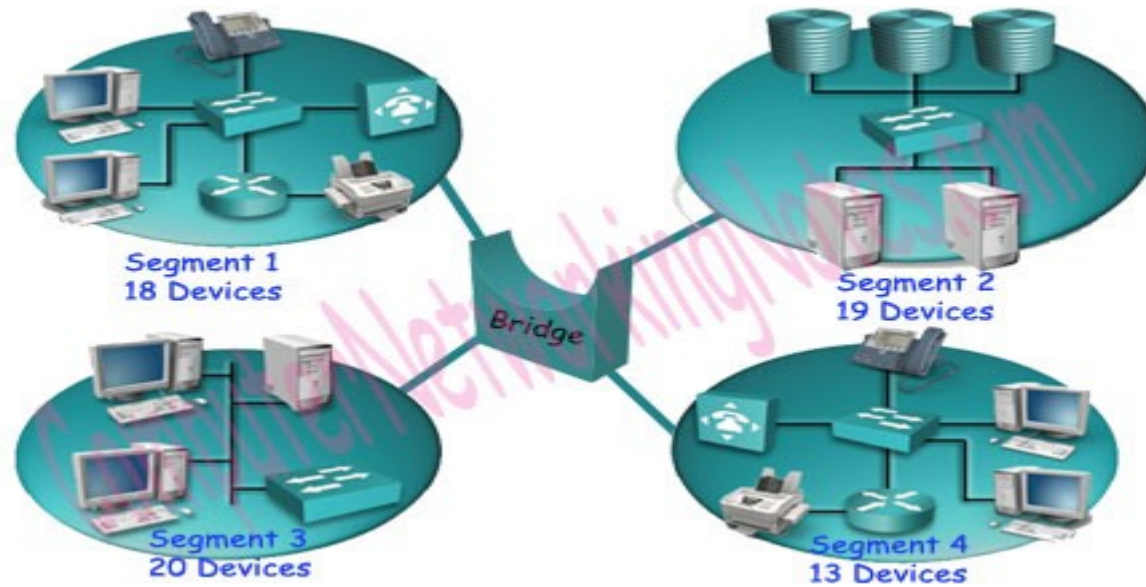
HUB



Bridge

Bridge is used to divide a large network into smaller segments. Basic functions of the Bridge are the following: -

- Breaking a large network into smaller segments.
- Connecting different media types. Such as connects UTP with the fiber optic.
- Connecting different network architectures. Such as connects Ethernet with the Token ring.



Switch

Just like Hub and Bridge, Switch is also used to connect the multiple devices together in a LAN segment.

Basically, a Switch is the upgraded version of the Bridge. Besides providing all the functionalities of Bridge, it also offers several additional features.

The biggest advantage of Switch is that, it makes switching decisions in hardware by using application specific integrated circuits (ASICs).

Unlike the generic processors that we use in our PC, ASICs are the specialized processors built only to perform very few particular tasks. Usually, the ASICs in the Switches have single task and that is the switching the frames as fast as possible.



Router

The router connects the different network segments.

It switches the data packets between those networks which are either located in the different logical segments or built with the different network layer protocols.

When a router receives a data packet on any of its interface, it checks the destination address of that packet and based on that destination address, it forwards that data packet from the interface which is connected with the destination address.

To forward a data packet to its destination, router keeps the records of connected networks. These records are maintained in a database table known as the routing table. Routing table can be built statically or dynamically.

Router

Basically routers are used: -

To connect different network segments.

To connect different network protocols such as IP and IPX.

To connect several smaller networks into a large network (known as the Internetwork)

To break a large network into smaller networks (Known as the Subnets. Usually created to improve the performance or manageability.)

To connect two different media types such as UTP and Fiber optical.

To connect two different network architectures such as Token Ring and Ethernet.

To connect LAN network with Telco company' office (Known as the DTE device).

To access DSL services (known as the DSL Router).

Router

